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proudly serving our community  
EMERGENCY MEDICAL SERVICE  
PROTOCOLS  
2026 REVISION V3  
08/17 /2026

7/25/2023

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## PROFESSIONALISM

EMS is really all about service. People like us, who have chosen to serve in EMS, have a moral and ethical obligation to behave in ways that serve the needs of others and reflect positively upon our organization, our community, and the whole discipline of EMS. This is not easy. In fact, it's a constant struggle that takes significant time, dedication, and effort. The choices we make and the manner in which we behave are a constant reflection of our professionalism. The following paragraphs were taken from Dr.

Brain Bledsoe's "Paramedic Emergency Care Second Edition," and establish the foundation that all

providers (volunteer or career) should strive to achieve and maintain.

"Professionalism describes the conduct or qualities that characterize a practitioner in a particular

field or occupation. Health care professionals promote quality patient care and take pride in their

profession. They earn the respect and confidence of team members by performing their duties to

the best of their abilities and by exhibiting a high level of respect for their profession.

Attaining

professionalism is not easy. It requires an understanding of what distinguishes the professional

from the non-professional. To develop this skill, keep the following points in mind.

Professionals

place the patient first; non-professionals place their egos first. Professionals practice their skills

to the point of mastery, and then keep practicing them to improve and remain sharp.

Nonprofessionals do not believe their skills will fade and see no reason to constantly strive for

improvement. Professionals understand the importance of response times; non-professionals get

to an accident when it's convenient. Professionals take refresher courses seriously, because they

know they have forgotten a lot and because they are eager for new information.

Non-professionals

believe they don't need training sessions and dislike being required to attend them.

Professionals

critically review their performance, always seeking a way to improve. Non-professionals look to

protect themselves, to hide inadequacies, and to place blame on others. Professionals check out

their equipment prior to the emergency response. Non-professionals hope that everything will work, supplies will be in place, batteries will be charged, and oxygen levels will be adequate.

Maintaining professionalism requires effort. But, the result of that effort - the admiration and

respect of one's peers - is the highest compliment a person can receive."

The NHTSA's National Standard Curriculum lists the following as professional behaviors:

- \* Integrity

- \* Empathy

- \* Self-motivation

- \* Appearance and personal hygiene

- \* Self-confidence

- \* Communications

- \* Time management

- \* Teamwork and diplomacy

- \* Respect

- \* Patient advocacy

- \* Careful delivery of services.

Being a professional has nothing to do with pay or rank or the level of certification that you hold. It is

the goal that every member of our practice, from EMT-Basic to the Medical Director, to constantly

strives for and remain a comprehensive, clinically sophisticated, and compassionate EMS

8/17 /2026

#### PROTOCOL AUTHORIZATION

The Gladstone Fire Department has developed the following set of medical protocols. These protocols have been reviewed and approved for use by Dr. Jared White, who serves as Medical Director.

#### Medical Directors Statement of Approval and Compliance

It is to our knowledge, all Gladstone Fire Department personnel meet or exceed the requirements as prescribed by the Missouri Bureau of EMS, and are licensed in good standing as required through policy, protocol, continuing education and skills competency evaluation.

I have reviewed and approve the medical protocols for the Gladstone Fire Department.

Approval Date: 08/17 /2026

Revision Date: 08/17 /2026

On file

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Dr. Jared White, Medical Director

---

Mike Desautels, Fire Chief

On file

05/23/2025

#### SYSTEMS DESIGN

The City of Gladstone shall provide the Gladstone Fire Department an Advanced Life Support (ALS) ambulance service. This service shall be ordered on calls for service within the City of Gladstone, as well requests from outside agencies for mutual aid. This service shall provide for a timely response, quality patient care and patient transports to an appropriate medical facility. In the event the fire department is unable to respond, a mutual aid provider shall be requested.

Non-emergency response shall include but not necessarily limited to the following:

1. Scheduled direct admissions to a hospital.
2. Hospital to home transfers, or nursing home to extended care facility transfers.
3. Home to hospital for medical tests.

Emergency responses shall include but not limited to the following:

1. Any call involving trauma resulting from an accident, fall, sports injury, burn, or etc., incident that results in injury requiring immediate medical attention.
2. Any call involving an illness resulting from diabetes, heart condition, strokes, seizures, or etc., any serious illness requiring immediate medical attention.

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#### MEDICAL CONTROL FORMAT

All patient treatment and transport shall follow guidelines as listed in current protocol.

Medical Control shall be contacted when indicated in the individual protocol, or when any problem or question exists. This is to ensure quality patient care on-scene, enroute to and upon arrival at the destination hospital.

When transporting a patient to a hospital other than the medical control hospital and orders are to be requested, the "other" receiving hospital should be contacted directly.

The following information should be transmitted to the medical control hospital in all emergency situations:

1. Paramedic's name and unit/service.
2. Expected arrival time.
3. Patient's chief complaint and brief history.
4. Patient's condition (including GCS).
5. Patient's vital signs.
6. Findings of exam and treatment.
7. Request for additional orders or questions.

05/23/2025

#### MEDICAL CONTROL

North Kansas City Hospital (NKCH) shall serve as the primary medical control hospital for the Gladstone Fire Department. All references within medical protocols unless otherwise stated, shall be directed toward North Kansas City Hospital.

Liberty Hospital (LH) shall serve as a secondary medical control hospital for the City of Gladstone, emergency medical service. If unable to contact North Kansas City Hospital, requests for orders shall be obtained from Liberty Hospital.

If the patient requests transport to an alternate hospital, medical control at the receiving hospital should be contacted for orders. If unable to obtain orders from the receiving facility, contact North Kansas City Hospital medical control for orders.

For Pediatric patients (under the age of 18), Children's Mercy Hospital (CMH) shall serve as the primary medical control hospital.



05/23/2025

#### DRUG ACCOUNTABILITY

The purpose of this protocol is to provide guidelines for the security, inventory and record keeping procedures for all scheduled narcotic medications in accordance with Fire / EMS Division policy F458 - Schedule Drug.

1. The following are considered scheduled narcotic drugs requiring strict accountability:

- A. Morphine Sulfate
- B. Versed - 2mg vials
- C. Versed - 5mg vials - Nasal
- D. Fentanyl - 100 mcg vials
- E. Ketamine - 500mg vials

2. The drugs listed above shall be kept in the following manner:

- A. Limited quantities shall be maintained in a sealed/tagged tamper proof container. This container shall be stored inside the Narcotic Safe located within the patient compartment of the ambulance.
- B. The following shall be considered minimum quantities to be carried in the ambulance at all times. Supply shall be restocked as soon as possible.

- 1. 4 Morphine Sulfate
- 2. 4 Versed 2mg
- 3. 4 Versed 5mg - Nasal
- 4. 4 Fentanyl - 100mcg
- 5. 2 Ketamine vials - 500mg

3. Daily inventory and drug accountability shall be the responsibility of the off-going and on-coming paramedic. Inventory and accountability shall be completed in the following manner:

A. At the beginning of each shift the off going and the on coming paramedics shall conduct a joint inventory of the scheduled drugs on the ambulance.

B. As the inventory is completed, each paramedic shall sign the logbook.

This log shall contain the following:

- 1. Date and time
- 2. Quantity of drug
- 3. Lock identification #
- 4. Paramedics' signatures and employee #

C. The inventory procedure shall be completed at any time there is a change in the on duty paramedic regardless of the time of day.

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D. On the 1st day of every month in the presence of an authorized officer the on duty paramedic shall open the sealed/tagged narcotic box and perform a physical inventory; also inspecting for signs of tampering and expired narcotics (see below). The unit container shall be restocked / resealed and documented in the logbook accordingly.

Broken Seal Procedure:

A. If a seal is found to be broken, the paramedic is to immediately notify a supervisor.

B. The supervisor and paramedic will together conduct an audit of the narcotics box and verify the contents.

C. The supervisor will make an entry in the narcotic log to document the broken seal; both the supervisor and medic will sign the log.

D. The supervisor will draft a memo to the Fire Chief going through the Chain of Command for his/her shift.(CC Narcotics Officer) explaining the incident.

E. At the discretion of the Fire Chief, a sample of a narcotic(s) may be sent to the Missouri State Highway Patrol lab for content verification.

Expired Medication Procedure

A. In the event expired narcotics are found during off hours, they shall be taped and marked as "expired do not use" placed back in the tagged narcotic container until they can be placed into the Supply Safe expired inventory.

B. In the presence of an authorized officer the expired narcotics will be removed and deducted from the ambulance inventory and noted as placed into the expired inventory.

C. The ambulance inventory shall then be restocked, re-tagged and logged, a piece of tape will be placed on the outside of the container with the name of the medications and expiration dates clearly listed.

D. Expired narcotics will then be placed into the expired container located in the Supply Safe and logged in the "Expired Narcotics Log," this is located in the back of the Supply Safe log. The expired container shall be re-tagged, logged and resecured.

4. Drug administration records shall be maintained on all scheduled drugs. These records shall be completed at the time of care and shall indicate the following:

A. Date and time of administration.

B. Name of patient.

C. Physician giving order for administration if applicable.

D. Quantity administered.

E. Paramedic's signature.

F. All unused portions shall be discarded at the hospital prior to returning to service. The amount discarded and the signature of the witness must be on the usage form. It is preferred HOSPITAL STAFF WITNESS DISPOSAL, If not possible have a supervisor witness.

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## HELICOPTER UTILIZATION

### 1. Indications

- A. Need for optimum speed of response and transport. (Keep in mind helicopter response and loading time).
- B. Impaired ground accessibility due to location, traffic or condition of roads.
- C. Shortage of ground transport units and there is a need for rapid transit.
- D. Need to achieve distribution of patients among area hospitals. (Mass casualty Incident).

### 2. Procedures

- A. It is the responsibility of the attending Paramedic and the Incident Commander to assess the situation and determine the need for air ambulance transport.
- B. If an air ambulance is needed it is the responsibility of the Incident Commander to contact dispatch and request it.
- C. It shall be the responsibility of the Incident Commander or designee along with law enforcement to establish a suitable landing site. This site should be:
  - 1. 100 feet square.
  - 2. Free of overhead obstructions.
  - 3. Accessible to aircraft for patient transfer.
  - 4. A fire apparatus with foam application capabilities should be available.

08/05/2026

## PATIENT DESTINATION CRITERIA

### 1. General

A. For all life-threatening emergencies transport the patient to the nearest appropriate facility.

B. When the nearest appropriate facility conflicts with the choice of the patient or a person acting in the behalf of the patient, contact Medical Control.

C. For all non-life-threatening emergencies, transport may be made to the following facilities: North Kansas City Hospital, St. Luke's Northland, Liberty Hospital, Truman/Truman West, Veterans Medical Center- Kansas City, St. Luke's-Plaza, K.U. Medical Center, and Research.

### 2. Trauma Criteria

A. Any trauma patient meeting the following criteria should be considered for transport to a Level I/II trauma center (NKCH, Liberty, KUMC or Truman)(Children's Mercy Hospital for patients under the age of 18) even if not the nearest facility, a high index of suspicion should be maintained regardless of time lapsed from point of injury to EMS patient contact:

#### Trauma Activation

-Trauma code

-GCS < 8 with MOI or other signs of shock.

-Confirmed & sustained systolic BP <90 & age specific for pediatrics.

-Respiratory compromise/obstruction and/or intubation Resp rate >29 or <10.

-GSW to head, neck, or torso.

-BP <90, Absent peripheral pulses.

-Fall with head injury of patient taking anticoagulants.

#### Trauma Alert

-Confirmed active hemorrhage

-GCS <14

-Extremity trauma with loss of distal pulse or sensation.

-MVC with:

Major auto deformity >1 foot

Unrestrained rollover

Death in same passenger compartment

Extrication >20 minutes

Ejection from vehicle

-MCC or ATV crash >20 mph or separation of rider from vehicle.

-Auto vs. pedestrian/bicycle

-Penetrating injury (i.e. stab wound) to head, neck, or torso.

-Major burns >20% BSA or involving inhalation injury.

-Falls > 3 feet.

-Pelvic fractures

-Open fractures

-Pregnant trauma patients >20 weeks gestation

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PATIENT DESTINATION CRITERIA CONT:

B. If NKCH, Liberty, and Truman are all on diversion, consider KU or St. Luke's on the Plaza for patients meeting criteria.

C. Trauma codes should be taken to the nearest Level I/II trauma center whether the trauma center is on diversion status or not.

D. Patients who meet the trauma routing criteria and are less than 18 years of age shall be transported to a pediatric trauma center (Children's Mercy Hospital).

HOSPITAL TCD (Time Critical Diagnosis) DESIGNATION

LEVEL 1 TRAUMA CENTERS:

\*KU Medical Center (PRIMARY Burn Center and Hyperbaric Chamber)

\*St. Luke's - Plaza

\*Research Medical Center (Secondary Burn Center)

\*Truman Medical Center

LEVEL 2 TRAUMA CENTERS:

\*North Kansas City Hospital (Primary)

\*Liberty Hospital (Secondary)

ACUTE STEMI

\*North Kansas City Hospital (Primary)

\*St. Luke's - Plaza (Secondary)

\*KU Medical Center (Secondary)

\*Liberty Hospital (

\*St. Luke's Northland

ACUTE CVA

\*St. Luke's - Plaza (Primary)

\*KU Medical Center (Secondary)

\*Research Medical Center (Secondary)

\*\*St. Lukes is the definitive stroke treatment center, however if a pt refuses to go there then

they may be taken to a stroke center of their choice. If they wish to go to NKCH educate them that there may be the possibility they could be transferred out of if they have certain problems. \*\*

Note: Patients meeting routing criteria SHALL NOT be taken to undesignated hospitals w/o medical control (physician) approval and/or contact with the receiving physician.

Note: Patient preference and continuity of care should always be considered, for example the patient has received all cardiac care at KU Medical Center, it would be beneficial for the

patient to be transported to KU even though it is not the closest facility. When in doubt contact medical control.

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BIOHAZARD EXPOSURE PROTECTION

See Department SOG F-311

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#### PRE-HOSPITAL DEATH DETERMINATION

Patients who have obviously expired will not be resuscitated or transported by Gladstone EMS personnel if any of the following obvious signs of death are present:

##### Trauma Related

A. If there is any doubt the arrest state is traumatic or any question concerning any sign of life at the scene, then resuscitation should be initiated and Medical Control contacted.

B. A patient with obvious traumatic injuries not consistent with human life, for example decapitation, incineration, transection of thorax (hemicorpectomy), massive trauma with visible organ damage; who is asystolic (confirmed in three leads) or has no audible heart sounds and who has not been observed at the scene to possess any sign of life (verbal or motor response, respiratory effort, or palpable pulse) may be pronounced DOA.

C. If unable to begin CPR in a reasonable amount of time due to patient accessibility or location, contact Medical Control and notify them of patient's condition and circumstances that prohibit appropriate resuscitation.

D. Documentation of any of the above findings shall occur in the ambulance run report.

##### Medical Related

A. A patient in cardiopulmonary arrest due to medical causes may be pronounced DOA if they are asystolic (confirmed in three leads) or have no audible heart sounds and also have one of the following signs:

1. Rigor mortis
2. Dependent lividity
3. Decomposition

B. Rigor mortis and asystole are not reliable signs of death in the hypothermic. If hypothermia is a consideration, resuscitation should be initiated.

C. Generalized tissue decomposition is not compatible with life; the patient may be pronounced DOA without cardiac monitor application.

D. If rigor mortis, lividity, or tissue decomposition is present, medical control contact is not necessary.

E. Any of the above findings shall be documented in the ambulance run report.

For all emergency scenes where patient needs exceed available EMS resources, initial assessment and treatment shall be in accordance with an approved triage methodology.

## REPORT WRITING

Documentation is one of the most important tasks that we perform as pre-hospital providers. While there are many reasons to document, the main purpose of documentation is to provide an accurate, comprehensive, and permanent record of each patient's condition and the treatment rendered. Always remember if it's not written down it didn't happen.

Any pertinent notes and information should be left with the receiving hospital. This would include any cardiac strips in cases of a STEMI. Signatures from the receiving hospital should be captured electronically on current RMS System. The formal EMS report shall be completed prior to the end of shift.

Listed below is the minimum criterion required for Gladstone EMS reports.

A. How the call was received from dispatch, pre-arrival information, units responding location and mode of response.

B. Nature of the call on arrival.

C. How the patient was found, presenting signs, appearance and pertinent scene factors.

D. The patient's chief complaint.

E. HPI & PPMH

F. Two sets of vital signs. This includes: blood pressure, pulse rate, respiratory rate, and SpO2 and EtCO2 readings if applicable.

G. IV attempts, location, patency and who initiates.

H. Intubation attempts, ET tube location measured from the teeth in cm, and confirmation methods.

I. Medication administration, name, dose, route and results.

J. Procedures performed, by whom and results.

K. The patient's overall response to treatment and transport.

L. Orders requested/received, and given by whom.

M. Attending Paramedic shall end the report with their full name, title and state license number (P#).

All calls shall be documented in the first person, past tense and in chronological order, from the time of dispatch through return to service.

All reports shall be written utilizing proper grammar, spelling and only Department approved abbreviations.

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GFD RECOGNIZED ABBREVIATIONS

AAA Abdominal Aortic Aneurysm  
AIDS Acquired Immune Deficiency Syndrome  
AMI Acute myocardial infarction  
ACLS Advanced cardiac life support  
ALS Advanced life support  
ABC Airway, Breathing, Circulation  
AC or ACF Antecubital or Antecubital Fossa  
A+Ox3 Alert and oriented to person, place and time  
APGAR Appearance, Pulse, Grimace, Activity, Respiration  
ASA Aspirin  
ASAP As soon as possible  
A-FIB Atrial fibrillation  
BVM Bag valve mask  
BLS Basic life support  
Bicarb Sodium Bicarbonate  
BP Blood pressure  
BM Bowel movement  
CA Cancer  
CPR Cardiopulmonary resuscitation  
CATH Catheter/Catheterization  
cm Centimeter  
CNS Central nervous system  
CVA Cerebral vascular accident  
CSF Cerebrospinal fluid  
C-Collar Cervical collar  
C-Spine Cervical spine  
CP Chest pain  
COPD Chronic obstructive pulmonary disease  
C/O Complains of  
CHF Congestive heart failure  
CPAP Continuous positive airway pressure  
CABG Coronary artery bypass graft  
CAD Coronary artery disease  
DOA Dead on arrival  
DVT Deep vein thrombosis  
DTs Delirium tremors  
DM Diabetes mellitus  
DC Discontinue  
mg/dL Milligrams per Deciliter  
DO Doctor of Osteopathy  
DNR Do not resuscitate  
DX Diagnosis  
Gtts Drops  
EKG/ECG Electrocardiogram  
EMS Emergency medical services  
EMT Emergency medical technician  
EMT-P Emergency medical technician paramedic  
ER Emergency room  
ET Endotracheal  
EPI Epinephrine

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= Equal

EOA Esophageal obturator airway

ETA Estimated time of arrival

EtCO<sub>2</sub> End tidal CO<sub>2</sub>

ETOH Ethyl alcohol

EJ External jugular

FD Fire department

FX Fracture

GI Gastrointestinal

Ga Gauge

GU Genitourinary

G#P# Gravida # Para #

GYN Gynecologist

HX History

HPI History of present illness

HIV Human immunodeficiency virus

HTN Hypertension

ID Identification

ICU Intensive care unit

ICP Intracranial pressure

IM Intramuscular

IN Intranasal

IO Intraosseous

IV Intravenous

IVP IV Push

JVD Jugular vein distention

KED Kendrick extrication device

kg Kilogram

LR Lactated ringers

LMP Last menstrual period

< Less than

LOC Level of consciousness-loss of consciousness

LPM Liters per minute

L-Spine Lumbar spine

MD Medical doctor

MEDS Medications

mcg Micrograms

meq or mEq milliequivalent

MAST Military anti-shock trousers

mg Milligram

MI Myocardial infarction

ML/ml Milliliter

mm Millimeter

> More than

AM Morning

MVC Motor vehicle collision

NC Nasal cannula

NG Nasogastric

N&V Nausea and vomiting

NTG Nitroglycerin

NKA or NKDA No known allergies, No Known Drug Allergies

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None

NRB Non-rebreather mask

NS Normal saline

NSR Normal sinus rhythm

OB Obstetrics

PO Orally by mouth

oz Ounce

OD Overdose

O2 Oxygen

SpO2 Oxygen Saturation Percentage

PMH Past medical history

PT Patient

PID Pelvic inflammatory disease

PCN Penicillin

PASG Pneumatic anti-shock garment

PD Police department

+ Positive

KCL Potassium chloride

Lbs Pound

Lpm Liters per minute

PAC Premature atrial contraction

PJC Premature junctional contraction

PVC Premature ventricular contraction

RX Prescription

1 Primary, first

PERL Pupils equal, reactive to light

PO Per Oral

ROM Range of Motion

RN Registered nurse

RESP Respiration, respiratory

R/O Rule out

RSR Regular Sinus Rhythm

2 Second, secondary

SOA Shortness of air or breath

SOP Standard operating procedure

SQ Subcutaneous

SL Sublingual

SIDS Sudden infant death syndrome

SIVP Slow IV Push

SVT Supra ventricular tachycardia

3 Tertiary, third

X Times

TKO To keep open

TIA Transient ischemic attack

TX Treatment

TB Tuberculosis

URI Upper respiratory infection

UTI Urinary tract infection

VD Venereal disease

V-FIB Ventricular fibrillation

V-TACH Ventricular tachycardia

WT Weight

Y/O Years old

D5W 5% dextrose in water

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#### PRE-HOSPITAL TERMINATION OF RESUSCITATION

Numerous studies have proven that a certain population of cardiac arrest patients will not survive hospital discharge if they fail pre-hospital therapy. Patients that meet the criteria

below shall be eligible for termination of resuscitative efforts in the field and do not require

transport to an Emergency Department. If there is any doubt, continue resuscitative efforts and transport.

##### Inclusion Criteria

- A. Primary cardiac arrest.
- B. Age >21 years of age.
- C. Either asystole or PEA as presenting rhythm.
- D. Asystole must be the terminal rhythm.
- E. ETT intubation.
- F. IV or IO access.
- G. > 20 minutes of appropriate therapy.
- H. Capnography reading less than 10 mm/Hg

##### Exclusion Criteria

A. Non-primary cardiac arrest (cause of arrest possibly attributed to other pathology). Including but not limited to the following:

- 1. Hypothermia
- 2. Drug overdose (possible acute overdose).
- 3. Primary respiratory arrest (i.e. status asthmaticus).
- 4. Carbon monoxide poisoning
- 5. Traumatic arrest.
- 6. Drowning
- B. Ventricular fibrillation or ventricular tachycardia at any time during resuscitation.
- C. Any return of spontaneous circulation during resuscitation.
- D. Any neurologic activity.
- E. Family desires transport (either voiced or perceived).
- F. Other, such as unstable scene or disagreement among family members.
- G. EMS responder witnessed cardiac arrest.

##### Procedure

A. Follow appropriate cardiac arrest protocol to include:

- 1. Successful early intubation.
- 2. Successful early IV drug therapy.
- 3. Minimum of 20 minutes of appropriate therapy.
- B. If the patient fits the inclusion criteria and does not fit any exclusion criteria and the above procedures have been accomplished the paramedic may initiate the termination of resuscitation.
- C. Paramedic does not need to contact medical control if all criteria is followed.
- D. The appropriate law enforcement agency must be notified.
- E. Rhythm strips must accompany report and report must contain the all proper documentation.

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#### PHYSICIAN ON SCENE

If a physician is found in attendance of a patient at the scene (excluding a physicians office),

or a physician comes to the scene after the paramedic has initiated treatment, the paramedic will:

- A. Advise the physician of your treatment, medical protocols, and additional directives received from Medical Control while continuing patient care.
- B. If a physician on scene desires to give medical direction regarding the patient care, the paramedic is to explain to the physician that you are functioning under strict medical control protocols. In order for the physician to assume patient care, the physician must be willing to assume full responsibility and liability for the patient. This includes accompanying the patient to the hospital. Medical Control must be contacted immediately and made aware of the situation. Ultimate determination of physician shall be made through Medical Control.
- C. If a physician in their office is treating a patient and orders are given, they shall be relayed to Medical Control. If a conflict in orders should arise, Medical Control shall overrule all other orders.

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#### QUALITY IMPROVEMENT PROCEDURE

The Quality Assurance/ Quality Improvement (QA/QI) Program is a systematic evaluation to ensure excellence in delivery of patient care and operations of the Gladstone EMS system.

Quality Assurance is a formal methodology designed to assess the quality of services provided and includes formal review of care, problem identification, corrective actions to remedy any deficiencies and evaluation of actions taken.

Quality Improvement is a continuous process that identifies problems in health care delivery, examines solutions to those problems, and regularly monitors the solutions for improvement. Our program will focus on ways to improve pre-hospital care rather than administering discipline and destructive criticism.

- A. To continually provide suggestions for improvement in the quality of medical care rendered.
- B. To improve patient care by adjusting protocols, treatment decisions, and procedures as needed.
- C. To ensure proper medical documentation.

#### Procedure

A. The EMS Quality Assurance Manager, Medical Director or his designee shall review the following categories of calls.

- 1. All cardiac arrests.
- 2. Any call marked life threatening.
- 3. All calls involving any new medication or procedure for the first six months of use.
- 4. Any call that varies from protocol.
- 5. Any call that the attending paramedic or supervisor desires reviewed.
- 6. Any STEMI activations.
- 7. Any Stroke activations.
- 8. Any Trauma activation.
- 9. Any Hospital diversions.
- 10. Academic review

B. Patient Safety Work Product (PSWP) meeting criteria for inclusion as defined in the Patient Safety Organization (PSO) Policy shall be promptly submitted to the Patient Safety Evaluation System (PSES) for evaluation and reporting to the Missouri Center for Patient Safety (MOCPS).

C. Upon the report return from the Medical Director, the attending Paramedic and Supervisors will review, comment (if needed) and sign the QA Worksheet.

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#### PARAMEDIC COMPETENCIES

Competencies are completed to ensure personnel are up to date on skills, service protocols and to familiarize employees with new equipment or procedures. This is just another way Gladstone Fire / EMS sets itself apart from other agencies by ensuring employees are highly trained and up-to-date with current equipment and procedures.

#### Minimum Performance Requirements (MPR's)

Bi-Annual Certification

Advanced Airway Management

Intubation (Nasal and Oral)

Invasive surgical airway

EtCO2 monitoring

CPAP Operation including in-line nebulizer Tx.

IV Access

IV ACCESS Proficiency - Difficult access

Must maintain a 70% success rate

Needle Decompression of the Chest

Cardiovascular Skills

EKG/ACS injury pattern

BLS

ACLS MegaCode

- AutoPulse Operation

- Cardiac Monitor Operations

Continuous Infusion Medications

Magnesium Sulfate

Cardizem

Epinephrine

Amiodarone

Bi-Annual Written Protocol Examination

Minimum Performance Standards (MPS) include application of knowledge / skills to approved protocols. This includes trauma, medical, environmental, cardiac arrest - adult/pediatric, scene management and any procedure used within the approved EMS protocols.

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#### ADULT/CHILD ABUSE REPORTING

Includes physical abuse, emotional abuse, sexual abuse, or neglect.

##### Duty to Report

Any person having reasonable cause to suspect that any child, a handicapped person, or elderly adult has been or may be abused or neglected shall make a report to the appropriate law enforcement agency and contact the following agencies:

Missouri Child Abuse Hotline - 1-800-392-3738

Missouri Adult Abuse Hotline - 1-800-392-0210

##### Assessment/Treatment

A. Provide necessary medical care according to the appropriate protocol.

B. Although a preliminary assessment of abuse or neglect is made in the field, appropriate personnel make the final determination upon review of the incident.

Accusations of abuse or neglect should not be made in the field.

##### Notification

A. In a private and confidential environment notify the physician at the receiving hospital of the facts upon which neglect or abuse is suspected.

B. Notification of the shift supervisor of the facts of the neglect or abuse is encouraged.

##### Documentation

A. The following information shall be included in the attending paramedics report:

1. All facts upon which abuse or neglect is suspected.
2. Identity of physician notified.
3. Whether physician or hospital staff is going to make proper notification.
4. If notification is made by the attending paramedic.



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#### PARENT-CHILD CONSIDERATIONS

- A. In general, a parent should accompany most conscious children during transport unless the parent's presence interferes with the care of the child.
- B. In situations of major trauma and cardiopulmonary arrest the paramedic should exercise his or her judgment as to the status of the parent riding in the ambulance.
- C. Allow the parent to hold and touch the child whenever possible.
- D. Both the parent and child respond best to open and honest dialogue.

05/23/2025

## Do-Not-Resuscitate / Advanced Directives / Emergency Order

### Sets

Missouri law provides for the observance of: Advance Directives, Do-Not-Resuscitate orders and

Emergency Order Sets when executed in compliance with Missouri state statute.

### Advanced Directives

The Advanced Directive becomes effective only when the declarant's condition is terminal and the

declarant does not have the capacity to make health-care decisions. Sec. 459.025, RSMo. It is

designed to allow a patient to forgo "death prolonging procedures," which are defined as "any

medical procedure or intervention which, when applied to a patient, would serve only to prolong the

dying process and where, in the judgment of the attending physician pursuant to usual and customary medical standards, death will occur within a short time whether or not such

procedure or

intervention is utilized. Death-prolonging procedures shall not include the administration of

medication or the performance of medical procedures deemed necessary to provide comfort, care

or to alleviate pain nor the performance of any procedure to provide nutrition or hydration." Sec.

459.010(3), RSMo.

### Transportable Physician Orders for Patient Preferences (TPOPP)

The Transportable Physician Orders for Patient Preferences (TPOPP) aims to improve the communication of a person's preferences regarding life-sustaining treatments and, thereby, generate high quality medical care in accordance with Sec. 459.010(3), RSMo. In order to accomplish this aim, TPOPP was designed to:

- \* Document a person's treatment preferences regarding CPR and the use of mechanical intervention for breathing/ventilation along with other life-sustaining treatments such as tube feedings, etc.

- \* Translate those preferences into an actionable portable set of physician orders

- \* Communicate an individual's care preferences across health settings

- \* Reduce repetitive documentation while complying with state laws and the Federal Patient Self-Determination Act.

When properly executed in accordance with Sec. 459.025, RSMo, the TPOPP shall be accepted and honored by the attending paramedic.

### Do-Not-Resuscitate (DNR)

Outside the hospital "Do-Not-Resuscitate" (DNR) requests are orders provided by a patient's physician to refrain from initiating cardiopulmonary resuscitative measures in the event of a cardiac

or respiratory arrest. DNR requests are compatible with maximal therapeutic care and the patient

may receive vigorous support (i.e. IV's & drugs) until the point of cardiac or respiratory arrest.

Outside the hospital "Do-Not-Resuscitate" orders, are regulated by RSMo sections 190.600 to 190.621. When properly executed, the DNR shall be accepted and honored by the attending paramedic. Missouri Bureau of EMS has ruled a photocopy / fax is also acceptable. DNR FAQ's

05/23/2025

#### Guidelines

Guidelines for acceptance of a DNR request include:

- A. If any doubt exists about the validity of the DNR form or the intent of the patient or the appropriate surrogate, resuscitation must be initiated and Medical Control contacted.
- B. Only the approved Bureau of EMS DNR form (MO580-1936) shall be used.
- C. The DNR form must be signed and dated by the patient or appropriate surrogate, a witness, and the patient's physician.
- D. The revocation provision must remain unsigned in order for this DNR form to remain valid.
- E. The DNR form must be with the patient at the time of the EMS providers' arrival.
- F. The EMS provider must see the form.
- G. No cardiac monitoring is necessary. No Medical Control contact is necessary.
- H. If this form is presented after care has begun, all resuscitative efforts shall cease and Medical Control shall be notified.

#### Documentation

The EMS report shall include the existence of a (DNR, DPA, TPOPP or other forms of Advanced Directives). A copy shall be attached to the EMS report. The original form shall remain at the

facility or home. If the patient is transported, the form(s) will accompany the patient.

#### Emergency Order Sets

##### Pediatric Emergency Form

Children's Mercy Hospital (CMH) has developed a form for special medical needs pediatric patients

who require very specific emergency medical care.

Upon arrival of EMS providers, the parent / caregiver shall provide a "Pediatric Emergency Form" to

EMS personnel, which contains detailed emergency instructions, medical history and hospital emergency contact information. The Pediatric Emergency Form shall supercede existing protocol and shall be followed by the attending paramedic.

If questions arise regarding the specific orders listed on the form, Children's Mercy Hospital can be

contacted at the phone numbers listed on the Pediatric Emergency Form. When contacted, the attending paramedic shall provide the hospital with the Medical Record (MR) number listed on the

top right corner of the form. Children's Mercy Hospital personnel will utilize the MR number to

confirm the specific orders.

In addition, CMH has advised parents / caregivers to have a "File of Life" information card located

on the refrigerator in the patients' residence.

#### Documentation

The EMS report shall include the existence of a "Pediatric Emergency Form." A copy shall be attached to the EMS report. The original form(s) shall be transported with the patient and given to

CMH staff.

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## PUBLIC SERVICE CALLS - DOCUMENTATION

### Purpose

To provide for adequate and uniform documentation of calls for public service or assistance.

These include but are not limited to:

- a. Lifting assistance.
- b. Assistance to or from a vehicle.
- c. Assistance with medical equipment.

### Procedure for documentation

- a. All public service calls shall be documented.
- b. These reports shall be completed using the NIFRS section of ESO.
- c. All reports shall include the patient's name and contact information
- d. The narrative shall be completed in accordance with EMS report standards (auto-generated reports are NOT acceptable forms of documentation).
- e. Any ambulance request that is dispatched with a nature other than Public Service requires the Gladstone Ambulance EMS Report be filled out along with the appropriate accompanying documents.

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#### ALS ENGINE REFUSALS

To allow Paramedics responding on ALS equipped fire apparatus to cancel an ambulance response and obtain a patient refusal.

##### Procedure

Gladstone ALS apparatus staffed by paramedic personnel may assess patient condition and need for ambulance transport. If it is determined, based on the assessment of the patient, a transport ambulance is not needed and/or the patient does not desire ambulance transport, the responding ambulance may be cancelled and a patient refusal accepted. It is the responsibility of the attending paramedic to explain all options for transport if medical attention is deemed necessary.

##### Documentation

Listed below is the minimum criterion required for Gladstone Fire Department reports.

How the call was received from dispatch, pre-arrival information, units responding location and mode of response:

1. Nature of the call on arrival.
2. How the patient was found, presenting signs, appearance and pertinent scene factors.
3. The patient's chief complaint.
4. HPI & PPMH
5. Vital signs. This includes: blood pressure, pulse rate, respiratory rate, and SpO2 and EtCO2 readings if applicable.
6. IV attempts, location, patency and who initiates.
7. Intubation attempts, ET tube location measured from the teeth in cm, and confirmation methods.
8. Medication administration, name, dose, route and results.
9. Procedures performed, by whom and results.
10. The patient's overall response to treatment.
11. Attending Paramedic shall end the report with their full name, title and state license number (P#).

All calls shall be documented in the first person, past tense and in chronological order, from

the time of dispatch through return to service.

All reports shall be written utilizing proper grammar, spelling and only Department approved abbreviations.

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## PATIENT REFUSALS

### Purpose

To establish guidelines and authorization for allowing a patient to refuse procedures, treatment and or transport.

### Procedure

The paramedic will be allowed to accept patient refusals provided sufficient information about the patient's condition have been given, allowing the patient or responsible person to make an informed decision, and:

1. Patient is a mentally competent adult, 18 years of age or older.
2. If the patient is not mentally competent, an authorized caregiver with power of attorney or guardianship may refuse in lieu of.
3. If the patient is under the age of 18, a parent or guardian must sign the refusal unless:
  - a. The minor is emancipated
  - b. Married
  - c. Pregnant
  - d. A Parent
  - e. A member of the armed forces
  - f. Financially independent, living away from home
4. In those incidents where the minor is 16 to 17 years of age and a parent or guardian is not available to sign a refusal, verbal consent by phone is acceptable provided sufficient information about the condition of the patient has been given, allowing the parent or guardian to make an informed decision. The name of the individual giving authorization for the refusal must be included in the narrative of the EMS report.
5. If the patient refuses care and/or transport and will not sign the refusal of care form, mark the form refused to sign, and document the refusal to sign in the narrative of the EMS report.
6. If there are any questions or problems regarding the appropriateness of a refusal, Medical Control shall be contacted.
7. All patient refusal reports shall be reviewed and initialed by the shift Captain and Battalion Chief prior to sending for final disposition.

05/23/2025

## ADVANCE BENEFICIARY NOTICE

### Purpose

To establish a guideline for the use of the Advanced Beneficiary Notice (ABN).

### Procedure

1. Medicare requires the Advanced Beneficiary Notice (ABN) when it is likely that costs for treatment and transport will not be covered.
2. The form is designed to help patients make an informed choice about whether or not to receive services, knowing that services may not be covered.
2. The form is to be filled out for all Medicare and Blue Cross insured patients.
3. The form is not intended to discourage treatment and/or transport; it is to help patients make an informed decision.
4. The attending paramedic shall inform Medicare and Blue Cross patients when it is possible that services will not be covered. The patient shall be given the form to read, check yes, services are wanted, or no, services are not desired. The form is to be signed and dated by the patient or a person acting on behalf of the patient.
5. Should the patient refuse to sign, date it and mark the form refused to sign.
6. The completed form shall be attached to the hospital face sheet and forwarded to the City billing clerk.

08/17/202

## CHEST PAIN/PRESSURE

Chest pain/pressure can be caused by a variety of different problems including: MI, angina pectoris,

pulmonary embolism, pneumonia and musculoskeletal chest wall pain.

Obtain a 12/15 lead EKG on chest pain of any kind, regardless if you think it's cardiac or not.

### STEMI SCENE TIMES SHOULD BE MINIMIZED

Secure Airway

Administer appropriate O2 concentration

Obtain IV access

Consider, ASPIRIN 324 mg to be chewed or swallowed if no history of asthma or sensitivity to aspirin.

Consider Nitroglycerin, as long as systolic BP remains >100 and pulse rate >50 and <140.

0.4 mg SL tablet every 5 minutes until chest pain is relieved to a total of 3 doses or hypotension develops.

-Consider Morphine if chest pain is unresolved with NTG, and systolic BP > 100.

-Give 4 mg initially, slow IV push, followed by 2 mg increments every 5 minutes until the pain is resolved or a total of 10 mg is given.

\*Consider Zofran 4mg SIVP or PO for nausea secondary Morphine administration.

\*Monitor BP, SpO2 and EtCO2 if indicated.

Treat arrhythmias per protocol or physician's orders.

Transport monitoring vital signs and cardiac rhythm frequently.

### Assessment

1. Obtain brief history characterizing pain as to the nature, severity, onset, duration, location, and radiation.

2. INITIAL 12/15 LEAD EKG SHOULD BE COMPLETED WITHIN 5 min OF PATIENT CONTACT

3. Obtain vitals, A BP SHOULD BE TAKEN ON BOTH ARMS, a difference of 15mmhg, systolic or diastolic and a history of hypertension could indicate an Aortic DISSECTION AND SHOULD BE REPORTED to receiving staff.

4. Perform a brief cardiopulmonary assessment to include inspection, palpation, and auscultation of the chest. Obtain current medications, past medical history and allergies.

If a STEMI is confirmed, refer to

Code STEMI

Activation

If there is no relief of chest pain after 1st dose of NTG,

you may administer Morphine.

\*WARNING\*

NITRATE ADMINISTRATION WITH SIGNS AND SYMPTOMS OF INFERIOR MI / RVI MAY CAUSE HEMODYNAMIC COLLAPSE.

V4R lead is the best lead for determining

RVI, with approximately 90 percent specificity.

Consider a fluid bolus and or use of vasopressors.



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CODE STEMI

MUST HAVES FOR ED ACTIVATION OF CODE STEMI:

\*Ongoing cardiac symptoms.

\*12/15 lead EKG show ST-elevation greater than or equal to 1 mm, in two or more Contiguous leads.

\*12/15 lead EKG without evidence of LBBB or Paced rhythm.

\*No obvious evidence of acute GI bleed or CVA.

STEMI SCENE TIMES SHOULD BE MINIMIZED, CONSIDER CODE 1

TRANSPORT WITH A SECOND PARAMEDIC.

INITIAL EKG POSITIVE FOR A STEMI,

\*CALL APPROVED CARDIAC CENTER AND PRE ALERT\*

PLACE PATIENT IN A GOWN

OBTAIN BILATERAL IV ACCESS IN THE ANTECUBITAL FOSSA

Consider Aspirin, 324 mg to be chewed or swallowed if no history of asthma or sensitivity to aspirin.

Consider Nitroglycerin, as long as systolic BP remains >100 and pulse rate >50 and <140.

0.4 mg SL spray or tablet every 5 minutes until chest pain is relieved to a total of 3 doses or hypotension develops.

If there's no relief of chest pain/pressure

after 1st dose of

NTG, you may

switch to

Morphine.

Treat arrhythmias per protocol or physician's orders.

CALL HOSPITAL IMMEDIATELY WITH A DETAILED

REPORT

Monitor vital signs and cardiac rhythm frequently.

Be sure to complete a GFD Field Note and leave a copy with the ER staff.

\*WARNING\*

AVOID

EXCESSIVE

VENIPUNCTURE

ATTEMPTS.

\*WARNING\*

NITRATE

ADMINISTRATION

WITH SIGNS AND

SYMPTOMS OF

INFERIOR / RVI

MAY CAUSE

HEMODYNAMIC

COLLAPSE.

V4R lead is the best

for determining RVI,

with approximately

90 percent

specificity.

Consider a fluid

bolus and or use

of vasopressors.

.

\*Consider Morphine if chest pain is unresolved with NTG, and systolic BP > 100.

\*Give 4 mg initially, slow IV push, followed by 2 mg increments every 5 minutes until the pain is resolved or a total of 10 mg.

\*Consider Zofran 4mg SIVP or PO for nausea secondary to Morphine administration.

\*Monitor BP, SpO2 and respiratory status closely.

Consider EtCO2

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## CARDIAC DYSRHYTHMIAS

### Unstable

A. Presence of a dysrhythmia with any of the following:

1. Altered mental status
2. Clinical signs of shock
3. Shortness of breath/pulmonary edema
4. Severe Chest pain/Pressure
5. Cyanosis

### Stable

A. Presence of a dysrhythmia and absence of above signs and symptoms.

B. Transport with supportive care only.

### General Treatment

A. Secure airway and assure ventilations.

B. Administer O2.

C. Start an IV. Consider an IO in severely obtunded children.

D. Monitor cardiac rhythm and vitals en route to hospital.

E. Proceed according to specific dysrhythmia protocol.

### Assessment

1. Obtain brief history of current symptoms, especially any chest pain or shortness of breath,

"tearing" back pain and/or past medical history.

2. INITIAL 12/15 LEAD EKG SHOULD BE COMPLETED WITHIN 5 min OF PATIENT CONTACT

3. Obtain vitals, A BP SHOULD BE TAKEN ON BOTH ARMS, a difference of 15 mmHg, systolic or diastolic and a history of hypertension could indicate an Aortic DISSECTION AND SHOULD BE REPORTED to receiving staff.

4. Briefly assess the cardiac and respiratory systems, and level of consciousness. Note any signs of poor perfusion such as altered level of consciousness, clammy skin, or hypotension.

7/25/2023

## NARROW COMPLEX TACHYDYSRHYTHMIAS

2

\*\*\*\*Identify and treat reversible causes\*\*\*\*

Is patient stable or unstable? 3

Unstable signs include a combinations of: altered mental status, ongoing severe chest pain/pressure, palpitations hypotension, SOA, diaphoresis.

Note: Rate related symptoms uncommon if heart rate < 150/min in adults and < 220 in children

STABLE 5

-Establish IV/IO access

-Review 12/15 lead ECG

UNSTABLE 4

-Perform immediate synchronized cardioversion: 50-100-150 J -Establish IV access and give sedation if patient is conscious; but do not delay cardioversion.

-If pulseless arrest develops, see pulseless arrest algorithm.

Is rhythm narrow

regular or irregular?

6

REGULAR 7

-Attempt vagal maneuvers

-Give Adenosine: 6 mg rapid IV push. If no conversion give 12 mg rapid IV push. May repeat 12 mg dose once.

STABLE IRREGULAR 8

You may consider:

Diltiazem for Afib/Aflutter with RVR, HR>120 AND:

1. Has a systolic BP of 100 or above.
  2. Symptomatic due to arrhythmia.
  3. Does NOT have a HX of WPW or CHF see note below.
  4. 12/15 Lead EKG confirms Afib/Flutter with RVR.
  5. New onset Afib less than 48hrs. Chronic Afib currently on Anticoagulants/Antiplatelet medications.
- Dosing: For patients over age 70 decrease dose by 5 mg.  
0.25 mg/kg max 20mg SIVP over 2 minutes, continue to monitor the PT, if hypotension develops, DC medication and consider a fluid bolus. Repeat 12 lead ECG if rhythm converts.  
If no response after 10 minutes, consider a second dose of 0.35mg/kg max 20mg SIVP over 2 minutes. If hypotension develops DC medication and consider a fluid bolus. Repeat 12 lead ECG if the rhythm converts.

Did the rhythm convert? 9

CONVERTS 10

If rhythm converts, probably

re-entry SVT:

-Observe for re-occurrence

-Obtain 12 lead EKG

DOES NOT CONVERT 11

Possible atrial flutter (see box 8) or

Junctional Tachycardia.

Assess and support ABC's 1

Give oxygen

Monitor ECG, blood pressure, SpO2 and EtCO2

Obtain 12/15 lead EKG

Obtain IV access, Antecubital preferred

General cardiac care if stable

Sedation: Versed

2-4mg IV initial

dose, subsequent

2mg to a max of

10mg

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WIDE COMPLEX TACHYDYSRHYTHMIAS

QRS > .12 seconds

Tachycardia with Pulses 1

-Assess and support ABC's 2

-Give oxygen

-Monitor EKG, blood pressure, SpO2 and EtCO2

-Identify and treat reversible causes

Is the patient stable? 3

Unstable signs include altered mental status, ongoing chest pain, hypotension, or other signs of shock.

Note: Rate related symptoms uncommon if heart rate

<150/min in adults and < 220 in children

STABLE 5

-Establish IV/IO access

-Obtain 12-lead EKG

UNSTABLE 4

-Perform immediate

synchronized cardioversion: 100-150-200 J

-Establish IV access and give sedation if patient is conscious; do not delay cardioversion.

-If pulseless arrest develops, see pulseless arrest algorithm. Is Rhythm regular? 6

REGULAR 7

If ventricular tachycardia or uncertain rhythm:

Amiodarone, 150 mg IV over 10 min.

-Prepare for elective synchronized cardioversion.

IRREGULAR 8

If atrial fibrillation with aberrancy

-See box 8 of narrow complex tachydysrhythmias.

If pre-excited atrial fibrillation (AF+WPW)

Avoid Adenosine

-Consider Amiodarone at 150 mg IV over 10 minutes

If Torsades de pointes

-Consider Magnesium 1-2 g over 5 minutes

PREMATURE VENTRICULAR CONTRACTIONS

Stable

Treat per general

protocol

Unstable

Unstable criteria + PVC's > 6/min, couplets, multifocal, or runs.

-Administer Amiodarone, 150 mg

IV over 10 min (15 mg/min). May

repeat same dose every 10 min

up to 2.2g total. Amiodarone drip mixed by placing

150 mg in a 100 ml bag of NS and

given over 10 minutes (15 mg/min).

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## BRADYCARDIA ALGORHYTHM

Bradycardia- Absolute/Relative?

Heart rate < 60 or inadequate

for clinical condition

-Maintain patent airway - assist breathing as

needed -Give oxygen

-Monitor ECG (identify rhythm , BP , SpO2 and EtCO2.

-Obtain 12-lead

-Establish IV access

Are their signs and symptoms of poor perfusion caused  
by bradycardia? (Acute altered mental status, severe  
chest pain/pressure, hypotension, etc.)

NO

Observe and Monitor

YES - Prepare for Pacing: use without delay for  
high-degree block (type II second degree or  
third-degree AV block)

-Consider Atropine (0.5 mg IV every 3-5 min)  
may repeat to a total dose of 3 mg or

0.04mg/kg. If ineffective, begin pacing.

- Consider - Epinephrine 1:100,000 (10-  
20mcg IVP every 2-3) if hypotension (SBP <  
90mmHg) persists after 500cc of NS.

- Identify and treat contributing causes

- Consider contacting medical control

Atropine, and

Epinephrine may be

ineffective with patients

who have overdosed on

beta-blockers.

05/23/2025

#### ASYSTOLE/PULSELESS ELECTRICAL ACTIVITY

Begin CPR/Autopulse 1

-Attach monitor and confirm Asystole or PEA

Monitor EtCO<sub>2</sub>

-Obtain IV or IO access 2

-Secure airway

-Administer Epinephrine 1 mg IV every 3

3-5 minutes.

Epinephrine 1:10,000

Epinephrine

may be given

via ET tube at

twice the dose

-Give 5 cycles of CPR 4

YES

Go to VF/VT

algorithm

NO -If no pulse go to box 3

-If pulse present, begin

post-resuscitation care

-Push hard and fast (100/min)

-Ensure full chest recoil

-Minimize interruptions in chest  
compressions.

-One cycle of CPR: 30 compressions then 2

breaths. 5 cycles = 2 minutes.

-Avoid hyperventilation.

-After a secure airway is placed and confirmed,  
rescuers no longer deliver "cycles."

-Give continuous chest compressions without  
pauses for breaths. Give 8-10 breaths/min.

Check rhythm every 2 minutes.

-Search for and treat possible

factors:

Hypovolemia

Hypoxia

Hypoglycemia

Hypo/Hyperkalemia

Acidosis

Toxins

Tamponade

Thrombosis

Trauma

Check rhythm.

Is it shockable?

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VENTRICULAR FIBRILLATION/PULSELESS VENTRICULAR  
TACHYCARDIA

If witnessed arrest, defibrillate first 1

Unwitnessed, CPR for 2 minutes

Check rhythm 2

Is it Shockable?

NO 8

-No pulse present, go to

Asystole/PEA algorithm

-If pulse present, begin

post-resuscitation care

YES 3

-Give 1 shock at 200 J

Resume CPR immediately

-Give 5 cycles of CPR 4

-Check rhythm

Is it Shockable?

YES 5

-Give 1 shock at 200 J

-Resume CPR immediately, regardless of rhythm  
change

-Administer Epinephrine 1 mg IV/IO every 3-5 minutes

-Give 5 cycles of CPR 6

-Check rhythm

Is it Shockable?

YES 7

-Give 1 shock at 200 J

-Resume CPR immediately

-Consider one: 1. Amiodarone: 300 mg IV/IO once.

May give an additional dose of 150  
mg once or

2. Lidocaine: 1 to 1.5 mg/kg initial dose.

Repeat at 0.5 to 0.75 mg/kg. Max 3  
doses or 3 mg/kg.

-If you have exhausted either antidysrhythmic with  
no response, you may switch to the other. Consider Double Sequential Defibrillation.

-Consider Magnesium: 1 to 2 g IV/IO for  
Torsades de pointes.

-Consider Sodium Bicarb: 1 meq/kg for  
prolonged down time.

-After 5 cycles of CPR, start sequence  
over at box 4.

-Push hard and fast (100/min)

-Ensure full chest recoil

-Minimize interruptions in chest  
compressions.

-One cycle of CPR: 30  
compressions then 2 breaths. 5  
cycles = 2 minutes.

-Avoid hyperventilation

-Secure airway and confirm.

-After an advanced airway is placed,  
rescuers no longer deliver "cycles"

-Give continuous chest  
compressions without pauses for  
breaths. Give 8 -10 breaths/min.  
Check rhythm every 2 minutes.

-

Epinephrine  
and Lidocaine  
can be given  
down the ET  
tube at twice  
the dose.

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POST RESUSCITATION (ROSC) PROTOCOL ADULT OR  
PEDIATRIC CARDIAC ARREST  
ASSESSMENT:

\*ROSC=pulse and measurable BP with increasing ETCO<sub>2</sub>

\* Continue to address specific differentials associated with original  
dysrhythmia

\*Focus on prevention of hypoxia, hypo/hypercapnia, hypotension(Survival  
and neurologic outcome worsen in their presence)

Optimize oxygenation/ventilation

ETCO<sub>2</sub> 35-45mmHg

DO NOT HYPERVENTILATE

Monitor vital signs and EKG rhythm

Obtain 12 lead ECG

SBP < 90 mmHg

NS BOLUS

ADULTS: 500mL IV/IO MAX 1L

PEDS: 20mL/kg REPEAT X 1

ADULTS: CONSIDER EPI 1:100,000 (PUSH DOSE)

1020mcg EVERY 23 MINUTES IF SBP < 90mmHg

IF SBP > 90mmHg AND ETT/King are In PLACE:

CONSIDER MIDAZOLAM FOR SEDATION: ADULTS: 12mg IV/

IO REPEAT AS NEEDED. MAX GIVEN 5mg PEDS: 0.1mg/kg UP

TO 2mg



05/23/2025

## RESPIRATORY DISTRESS

Respiratory distress can be divided into processes involving the upper or lower airway.

Examples of

upper airway obstruction include: foreign body, epiglottitis, croup, and trauma. Diseases that cause

lower airway obstruction include: asthma, COPD, pneumonia, CHF, and pneumothorax. If unable to

differentiate the cause of respiratory distress, open and maintain airway, administer oxygen, ensure

adequate ventilation, and transport rapidly.

Assessment:

1. Assess adequacy of airway and ventilation to include: vocalization, rate and depth, breathe sounds, cyanosis, and signs of respiratory distress.
2. Perform brief cardiovascular assessment.
3. Check vital signs to include SpO2 and EtCO2.
4. Obtain brief history to include: onset, presence of chest pain, potential aspiration, fever, and past medical history.

### Upper Airway Obstruction

1. Open airway and assess for breathing.
2. If no breathing present follow AHA guidelines for obstructed airway. Attempt ventilation.
3. If unsuccessful, use direct laryngoscopy to check for foreign bodies. If present, remove with magill forceps. Assist ventilations as needed.
4. If still unsuccessful, consider surgical cricothyrotomy airway or commercial cricothyrotomy device. All other airway techniques must be attempted prior to surgical airway.
5. If resolved or initially only partially obstructed, do the following:
  - a. Transport in upright position.
  - b. Administer O2.
  - c. Establish IV access.
  - d. Monitor respiratory status.

### Asthma

1. Ensure airway.
2. Establish IV access.
3. Administer O2 using appropriate administration set, for Status Asthmaticus consider early CPAP with inline breathing treatment. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider Versed 2-4 mg SIVP up to a total dose of 10 mg. If unable to obtain IV access, consider IN Versed 2 - 4 mg, max 10 mg. See intranasal procedure.
4. Duoneb 3 ml nebulized. Set O2 at 6-8 lpm. Consider if needed Albuterol 2.5 mg x 2, heart rate should not exceed 150 bpm. Xopenex may also be considered if other medications are ineffective or contra-indicated (Pediatric dosing of Xopenex, dilute with 3ml of NS)
5. Consider Solu-Medrol 62.5 mg IV. Pediatric 2mg/kg **USE WITH EXTREME CAUTION IN DIABETIC PATIENTS AS MAY INCREASE INSULIN METABOLIC REQUIREMENTS**
6. For severe cases consider Magnesium Sulfate 1g drip over 10 min, re evaluate recheck vitals and then give the second 1g dose over 10 min. Consider Epinephrine 1:1,000 concentration, 0.3 mg IM. Contact medical control for orders if age is over 40 AND pulse rate > 120, or HX of CAD. Consider Terbutaline 0.25mg SQ.
7. Consider BiPap.

Note: Surgical airway may be necessary in severe maxo-facial trauma as a first line intervention

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## RESPIRATORY DISTRESS (CONT)

Congestive Heart Failure- (see CHF Protocol)

1. Ensure airway.
2. CPAP for severe respiratory distress; Administer O2 and assist ventilations as needed. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider Versed 2-4 mg SIVP up to a total dose of 10 mg. If unable to obtain IV access, consider IN Versed 2-4 mg, max 10 mg. See intranasal procedure.
3. Monitor cardiac rhythm and obtain 12/15 lead ECG.
4. Start an IV at a TKO rate. Carefully monitor fluid.
5. Nitroglycerin 0.4mg SL if the patient maintains a systolic blood pressure > 100, and a pulse rate >50 and <140. May be repeated every 5 minutes until distress is relieved, or systolic blood pressure < 100, or 3 doses.
6. If EtCO2 indicates constrictive pattern you may consider Duoneb 3 ml nebulized. Set O2 at 6-8 Lpm. Consider if needed Albuterol 2.5mg x 2, heart rate should not exceeds 150 bpm. Xoponex may also be considered if other medications are ineffective or contra-indicated.
7. Transport monitoring vital signs.

Tension Pneumothorax

1. Secure airway.
2. Administer O2.
3. Monitor cardiac rhythm.
4. Needle decompression as per decompression procedure.
5. Transport monitoring vital signs and respiratory status.

COPD

1. Ensure airway.
2. Administer O2. Consider CPAP for severe respiratory distress. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider Versed 2-4 mg SIVP up to a total dose. If unable to obtain IV access, consider IN Versed 2-4 mg, max 10 mg. See intranasal procedure
3. Start an IV at a TKO rate.
4. Apply cardiac monitor and obtain 12/15 lead ECG.
5. Duoneb 3 ml nebulized. Set O2 at 6-8 Lpm.
6. Consider if needed Albuterol 2.5 mg x 2, heart rate should not exceeds 150 bpm. Xoponex may also be considered if other medications are ineffective or contra-indicated.
7. For severe cases consider Magnesium Sulfate 1g drip over 10min, re-evaluate recheck vitals and then give the second 1g dose over 4-5 minutes.
8. Consider Solu-Medrol 62.5 mg IV, Pediatric 2mg/kg. USE WITH EXTREME CAUTION IN DIABETIC PATIENTS AS MAY INCREASE INSULIN METOBOLIC REQUIREMENTS
9. Consider BiPap.

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## RESPIRATORY DISTRESS (CONT)

Pediatric Respiratory Distress with Inspiratory

Stridor/Croup

Treatment:

1. Humidified O2, as appropriate, with upright posture

2. If needed, assist ventilations with PPV using 100% O2

3. For signs of severe croup, administer inhalation of nebulized solution of Epinephrine 1:1,000 (1mg/mL) at the following dose:

Signs of severe croup include stridor at rest AND one of the following: tachypnea, moderate inter

costal retractions (including suprasternal retractions), agitation/restlessness/tired appearing, difficulty talking or feeding.

4. Contact Medical Control for the second dose of nebulized Epinephrine if symptoms do not improve consider alternate diagnosis such as aspiration of foreign body, bacterial tracheitis, epiglottitis.

5-10 kg: Nebulized Epinephrine Mix 0.5 mg (0.5 ml) Epi

1:1000 with 2ml of normal saline

>10 kg: Mix 1mg(1ml) Epi 1:1000 with 4ml of normal saline

5-10 kg: 2.5mL

Greater than 10kg: 5mL

\*Albuterol is not an appropriate treatment for croup

\* Nebulized Epinephrine may be contraindicated in children with a history of congenital heart disease.

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## ALLERGIC REACTION/ANAPHYLAXIS

Allergic reactions can be mild to life threatening. Potential routes for exposure include inhalation, ingestion, injection, or direct contact. Treatment addresses mild anaphylaxis, anaphylaxis with respiratory distress, and anaphylactic shock. If cardiopulmonary arrest should occur, proceed to the cardiac arrest protocol.

### Assessment

1. Check vital signs.
2. Obtain brief history with attention paid to known allergies or any recent exposures.
3. Assess cardiac and pulmonary systems.

-Secure Airway.

-Administer O2.

-Place on cardiac monitor, SpO2 and EtCO2.

-Establish IV access.

-Transport immediately

### Localized Allergic

#### Reaction

(normal BP & pulse with no stridor or respiratory distress)

Mild Anaphylaxis with

Respiratory Distress

Anaphylactic Shock

Consider Benadryl:

25-50 mg IV, Pediatric

1 mg/kg IV, up to a total

single dose of 50 mg.

### Transport

Duoneb 3 ml nebulized. Set O2 at

6-8 Lpm. Consider if needed

Albuterol 2.5mg x 2, heart rate

should not exceeds 150 bpm.

Give a fluid bolus. 20 ml/kg up to a

maximum of 500 ml. Consider a

repeat bolus if needed to maintain

blood pressure.

### SUPPORT RESPIRATORY,

#### CONSIDER BRONCHODILATORS

-Administration of Epinephrine 1:1,000

0.3 to 0.5mg, Sub Lingual injection or IM

-Preferred - venous plexus at the base

of the tongue.

-Peds 1:10,000 Epinephrine 0.01 mg/kg

IM

1:1,000 concentration - 0.3 to 0.5 ML = .3 - .5 mg

1:10,000 concentration - 3 to 5 ML = .3 - .5 mg

Contact medical control

for orders for Epinephrine

in any of the columns, for

patients over age 40 with

a HX of CAD.

For a pulse rate > 120.

You may consider

Terbutaline 0.25mg SQ

Supportive care, watch for

signs of decompensation

Xoponex may also be considered

if other medications are contraindicated or ineffective. Pediatric

dosing, dilute in 3ml of NS.

\*\*\*\*For CRITICAL Patients\*\*\*\*

Consider Epinephrine Drip 2-

10mcq/min, .5mg of Epi mixed in

500cc admin thru 60gtt set, drip at

2ml/min titrate to effect.

Consider SoluMedrol 62.5 mg

05/23/2025

## ABDOMINAL PAIN

### Assessment

1. Take vital signs.
2. Obtain brief history including onset, duration, and past medical history.
3. Briefly assess cardiac, pulmonary, and neurological systems.
4. Identify possible causes.
5. Consider gynecological emergencies in females. Refer to gynecological emergencies protocol.

-Administer oxygen as appropriate.

-Monitor cardiac rhythm.

-Establish an IV.

If patient is hypotensive,

Consider a 20ml/kg fluid bolus (max 1000ml)

if lung sounds are clear.

If nausea/vomiting present,

-Consider Zofran: 4 mg SIVP or PO

-Children (4 and older): 2 mg IV.

Must contact medical control for orders  
prior to administering to Infants.

Contact medical control for additional  
doses.

Contact medical control for orders and  
dose for children under 4 years of age.

- Consider Fentanyl 25-50mcg slow  
IV, IM or IN

-Consider Morphine 2-4mg IV

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## BEHAVIORAL/PSYCHIATRIC DISORDERS

### Assessment

1. Check vital signs.
2. Obtain brief history noting any bizarre or abrupt changes in behavior, suicidal intentions, alcohol or drug ingestion, and any significant past medical history.
3. Perform a physical exam if possible.
4. Perform a brief neurological exam including mental status.
5. If restraints are used a minimum of two people will accompany patient until care has been transferred.

Probable Psychiatric

Problem Only

Lethargic, Unstable Vital

Signs, or Hypoglycemia

Suspected

Restrain in lateral

recumbent, supine, or

prone position only if

absolutely necessary.

-Attempt to establish  
rapport with patient.

-If patient is dangerous to  
self or others, consider law  
enforcement assistance.

If suicidal, do not leave  
patient alone and if  
possible remove any  
dangerous objects.

If emergency transport is  
unnecessary, do as little as  
possible except reassure  
the patient during transport.

-Administer O2.

-Apply EKG

-Start an IV, TKO.

-Do a glucose check.

Agitated

See Excited Delirium  
protocol

If patient is

restrained,

continuously

monitor airway and

breathing.

Consider administration of  
Dextrose.

Adults: D10 250mL IV

Ped: >8 yrs - D10 2mL/kg IV

Ped: <8 yrs - D10 1mL/kg IV

Consider Narcan: 0.4-2 mg

SIVP, titrate to respiratory  
effort, monitor EtCO<sub>2</sub>, IM,

ET, IO, or IN.

Transport monitoring vital  
signs.

If trauma or dysrhythmia is  
noted proceed to  
appropriate protocol.

Consider own safety  
and limitations.

If patient is dangerous  
to self or others,  
request law  
enforcement  
assistance.

Consider Versed 5-10mg

Restrain in lateral recumbent,

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#### CONGESTIVE HEART FAILURE (CHF)

1. Ensure airway.
2. Rapid CPAP for severe respiratory distress; Administer O2 and assist ventilations as needed. Intubation should be considered in patients exhibiting signs of physical exhaustion/respiratory failure. If the patient is conscious and needs to be intubated, consider Versed 2-4 mg SIVP up to a total dose of 10mg. If unable to obtain IV access, consider IN Versed 2-4 mg, max 10 mg. See intranasal procedure.
3. Monitor cardiac rhythm and obtain 12/15 lead ECG.
4. Start an IV at a TKO rate. Carefully monitor fluid.
5. Nitroglycerin 0.4mg SL if the patient maintains a systolic blood pressure > 100, and a pulse rate >50 and <140. May be repeated every 5 minutes until distress is relieved, or systolic blood pressure < 100, or 3 doses.
6. If EtCO2 indicates constrictive pattern you may consider Duoneb 3 ml nebulized. Set O2 at 6-8 Lpm. Consider if needed Albuterol 2.5mg x 2, heart rate should not exceeds 150 bpm. Xoponex may also be considered if other medications are ineffective or contraindicated.









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## CVA/STROKES

Consider stroke in any patient presenting with the following:

- Altered mental status
- Dysarthria (slurred speech)
- Aphasia (difficulty understanding or expressing speech)
- Weakness or Numbness in the face, arm, or leg
- Vision (blurry vision, double vision, Blindness)
- Dizziness, loss of balance, loss of coordination
- Headache (especially if sudden or severe)

### Assessment

1. Check vital signs
2. Briefly assess pulmonary, cardiac, and neurological systems.
3. Determine time patient was last known normal
4. Administer Cincinnati Stroke Scale
5. Administer VAN Score
6. Determine Blood Glucose Level
7. Other specific history of importance includes
  - a. medications (especially anticoagulants),
  - b. recent surgeries (3 month brain or spine, 21 days major surgeries)
  - c. history of brain tumor or aneurysm, history of bleeding disorders.

### Treatment

1. Administer O2 as appropriate to maintain O2 saturation >94%
2. Place on Cardiac Monitor and pulse oximeter
3. Start an IV TKO
4. Treat blood glucose per protocol

05/23/2025

Van Assessment Tool for Large Vessel Occlusion:

HOW WEAK IS PATIENT

ON ONE SIDE OF THE

BODY?

VISUAL

DISTURBANCE?

APHASIA?

1

2

3

VAN

NEGLECT?

If patient has any weakness PLUS any one of the below:

Visual Disturbance (field cut, double, or blind vision)

Aphasia (inability to speak or understand)

Neglect (gaze to one side or ignoring one side)

This is likely a large artery clot (cortical symptoms) = VAN Positive

All VAN positive patients should be sent to endovascular capable hospital and notified ahead of time. NeuroIR paged with VAN positive patient arriving. CT/CTA done on arrival

4

Mild (minor drift)

Moderate (severe drift - touches or nearly touches ground)

Severe

No Weakness

Field cut

Double vision

Blind new onset

NONE

Expressive

Receptive

Mixed

NONE

Forced gazed or inability to track one side

Unable to feel both sides at same time or unable to identify own arm

Ignoring one side

NONE

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Stroke Routing Protocol:

If Van Negative:

Transport to the closest Primary Stroke  
Center (NKCH).

Notify ED Triage "This is a stroke alert"

If VAN Positive:

Transport to the closest Comprehensive  
Stroke Center (St Lukes Plaza, KUMC,  
Research). Notify ED Triage "This is a  
stroke alert with possible large vessel  
occlusion"

05/23/2025

## DEHYDRATION

### Assessment

1. Check vital signs; should include Orthostatic BP.
2. Obtain brief history noting onset, any complaints of pain, past medical history, and medications.

-Ensure airway.

-Administer O2 using appropriate administration set.

-Monitor cardiac rhythm.

-Obtain a glucose reading.

-Establish an IV.

- Consider a 20ml/kg fluid bolus (max 1000ml)

if lung sounds are clear.

-For pediatrics, consider 20ml/kg bolus.

Contact med control for additional orders if needed

Treat associated dysrhythmias as per protocol.

### Reassess

lung sounds

frequently

07/07/2026

## DIABETIC EMERGENCIES

### Assessment

1 Check vital signs.

2. Obtain brief history noting onset and progression of present state, past medical history, and medications.

3. Consider other etiologies (drug use, alcohol).

-Administer O2 as appropriate.

-Establish an IV.

-Consider 12/15 Lead EKG/  
monitor to r/o AMI with a positive  
cardiac history.

15g oral glucose may be considered for awake,  
cooperative patients with a patent airway and  
the ability to swallow.

Consider Glucagon if unable to obtain IV access.

Adult: 1 mg IM

Pediatric: Greater than 25kg - Give 1mg IM

Less than 25kg - Give 0.5mg IM

### MUST TRANSPORT AFTER ADMINISTERING.

Glucose checks should be done prior to and  
after administration of Dextrose.

Consider additional Dextrose administration if  
patient remains unconscious or disoriented.

### Hyperglycemia

Consider a 20 ml/kg NS fluid bolus (max  
1000ml) if blood glucose reading is over  
400 mg/dL.

-Use caution in CHF patients.

-Patients must maintain clear lung sounds.

### Hypoglycemia (< 60 mg/dL)

Transport monitoring vital signs.

Transport monitoring vital signs.

<http://circ.ahajournals.org>

/content/97/1/12

### Diabetes/Cardiac relation

Refer to Broselow tape for weight estimation  
as needed.

Adults: Give Dextrose - D10 250ml (25g) IV

Greater than 8 years of age: 2 mL/kg D10 IV

Less than 8 years of age: 1 mL/kg D10 IV

05/23/2025

#### HYPER-AGITATED DELIRIUM SYNDROME

The purpose of this protocol is to identify and treat those patients experiencing Hyper-Agitated Delirium by mitigating the effects and preventing sudden death.

#### TREATMENT:

- \* Request police assistance immediately.

- \* Initiate General Patient Care.

- \* Presentation:

Identify possible Hyper-Agitated Delirium using acronym PRIORITY. Presence of some or all of these symptoms is indicative of potential syndrome.

- a. P: Psychological issues

- b. R: Recent drug/alcohol use

- c. I: Incoherent thought processes

- d. O: Off (taking clothes off) and sweating, HYPERTHERMIA

- e. R: Resistant to presence or dialogue

- f. I: Inanimate objects: violent toward shiny or glass objects

- g. T: Tough, unstoppable, superhuman strength

- h. Y: Yelling

- \* Providers should ensure crew safety at all times. (See Restraint Protocol)

- \* Consider Ketamine 3-5 mg/kg IM. Maximum 500 mg. Contact Medical Control for further doses.

- \* Consider Midazolam "Versed"

- a. 5-10 mg IM/IN

- b. 5-10 mg IV, 2 mg increments

- c. 0.1 mg/kg IV (Pediatric dose)

- d. Maximum total dosing is 10 mg, call med control for additional

- \* Establish IV access after sedation, if not already established.

- a. Infuse 1L of cool (room temp, not from a warmer) Normal Saline

- b. Apply capnography to monitor patient respiratory status.

- \* Prepare to aggressively manage the airway if needed.

- \* Rapid Transport to the closest appropriate hospital with Medic and rider to assist.

#### Sudden Death Factors

- \* Metabolic Acidosis

- Potentially life threatening abnormality of the blood chemistry

- Aggravated by:

- \* Increased activity, violent confrontation, dehydration

- \* Blood acidity affects vital functions

- \* Elevated blood potassium level

- Survivors

- \* Kidney damage due to muscle breakdown

- \* May require dialysis



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## GYNECOLOGICAL EMERGENCIES

### Assessment

1. Take vital signs.
2. Obtain brief history including onset, last menstrual cycle, past medical history and amount of blood loss if any (number of pads used in what time frame).
3. Briefly assess cardiac, pulmonary, and neurological systems.
  - Administer oxygen as appropriate.
  - Monitor cardiac rhythm.
  - Establish an IV.

### Vaginal Bleeding

Consider a 20ml/kg fluid bolus  
(max 1000ml) if lung sounds are clear.

- Monitor lung sounds.

### Emergency Childbirth,

### Postpartum Care, or

### Postpartum Hemorrhage

Refer to childbirth protocol

Some causes of vaginal bleeding could include: ectopic pregnancy, miscarriage, abruptio placentae, placenta previa, and pre-term labor.

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## HYPOTENSION

### Assessment

1. Obtain vitals, A BP SHOULD BE TAKEN ON BOTH ARMS, a difference of 15mmhg, systolic or diastolic and a history of hypertension could indicate an Aortic Dissection and SHOULD BE REPORTED. Consider Orthostatic Vitals. Determine if absolute or relative.
2. Obtain brief history, BP norm and noting any complaints involving the cardiovascular and respiratory systems, onset, past medical history, and medications.
3. Consider obtaining an early EKG 12/15 lead.

-Ensure airway.

-Administer O2

Monitor cardiac rhythm.

-Start an IV of NS.

- Consider a 20ml/kg fluid bolus (max 1000ml)  
if lung sounds are clear.

Use caution if cardiac origin is suspected

-If cardiac origin, - Consider - Epinephrine  
1:100,000 (10- 20mcg IVP every 2-3 minutes) if  
hypotension (SBP < 90mmHg) persists after  
minimum of 500cc of NS.

-Transport.

-Monitor vital signs and cardiac rhythm.

Contact medical control for additional fluid  
bolus, ask for amount and rate.

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## **HYPERTHERMIA**

Hyperthermia may be the result of elevations in body temperature secondary to infections, drug ingestions, chronic diseases, and environmental conditions. Elevations in body temperature may induce seizures in children. Hyperthermia with an altered mental status is a true emergency regardless of the etiology.

### **Assessment**

1. Check vital signs including a temperature. Significant hyperthermia is present when core body temperature exceeds 105 F.

2. Obtain brief history noting length and type of exposure, exercise related, illness related, and environmental etiology.

3. Briefly perform cardiac, respiratory, and mental status assessments. Note the condition of the skin.

4. Remember that the elderly and pediatrics are more susceptible to heat.

Treatment without altered mental status      Treatment with altered mental status

-Loosen clothing or remove any heavy clothing.

-Do not cover with blankets.

-Secure airway.

-Administer O2 as needed.

Consider starting an IV, TKO or as directed.

Consider cooling the patient by sponging with room temperature water. Avoid inducing shivering.

Place on cardiac monitor if the patient has an abnormal pulse rate, respiratory rate, or blood pressure.

-Secure airway.

-Administer O2 as appropriate.

-Assist ventilations as needed.

-Establish an IV, TKO or as directed.

-Place on cardiac monitor.

\*Do a glucose check. Consider administration of Dextrose if indicated:

Adults: Give Dextrose - D10 250ml (25g) IV

Greater than 8 years of age: 2 mL/kg D10 IV

Younger than 8 years of age: 1 mL/kg D10 IV  
ml of D50 IV.

-Remove clothing and consider cooling by sponging patient with room temperature water, and if possible fan the patient to achieve evaporation.

-Avoid inducing shivering.

Consider administration of Narcan 0.4 -2 mg

SIVP, titrate to respiratory effort, monitor EtCO2.

Transport monitoring vital signs.

Do not delay transport to perform cooling in the field.

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## HYPOTHERMIA

Children in general and neonates in particular are more susceptible to hypothermia than adults.

### Assessment

1. Check vital signs including temperature. Severe hypothermia mimics death.
2. Obtain brief history noting length of exposure and pertinent medical history.
3. Briefly assess cardiac status, mental status, and skin for evidence of local injury such as frostbite.

-Handle the patient as gently as possible.

-Avoid any unnecessary manipulation.

-Administer O2 as appropriate.

-Assist ventilations as needed.

Do CPR as necessary.

Establish an IV, TKO or as directed.

Remove any wet clothing and wrap patient in blankets.

Transport while monitoring vital signs and cardiac rhythm.

Secure airway with as little manipulation as possible.

Stimulation of the airway can cause ventricular fibrillation.

Do a glucose check.

Consider administration of Dextrose if indicated:

Adults: Give Dextrose - D10 250ml (25g) IV

Greater than 8 years of age: 2 mL/kg D10 IV

Younger than 8 years of age: 1 mL/kg D10 IV

-Place patient on cardiac monitor.

-If patient is in VF, attempt defibrillation.

Consider administration of Narcan 0.4-2 mg SIVP, titrate to respiratory effort, monitor EtCO2.

### Local (Frostbite)

Handle injured part very

gently. Protect from

pressure, trauma, or further

environmental hazards.

Do not allow the limb to thaw

if there is a chance of

refreezing before evacuation

and transport is complete.

-Maintain core temperature

with blankets.

-Monitor vital signs.

Active rewarming should

occur at the hospital, not in

the field.

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#### NAUSEA / VOMITING

For the prevention of nausea and vomiting in acute medical conditions: including headache, chest pain, abdominal pain, chemotherapy, or post-operative induced nausea and vomiting.

##### Assessment

1. Take vital signs.
2. Obtain brief history including: onset, duration, and past medical history.
3. Briefly assess cardiac, pulmonary, and neurological systems.

Secure airway.

-Establish an IV.

Consider a 20ml/kg fluid bolus (max 1000ml) if blood pressure is <90 mm/hg.

-Lung sounds must be clear

-Consider Zofran: 4 mg SIVP or PO

-Children (4 and younger): 2 mg IV.

Must contact medical control for orders prior to administering to children.

Contact medical control for additional doses.

Contact medical control for orders and dose for children under 4 years of age.

05/23/2025

## NEAR DROWNING

### Assessment

1. Check vital signs. If pulseless and non-breathing, follow cardiopulmonary arrest protocol.

2. Obtain brief history noting estimated submersion time, water temperature, and mechanism of injury.

3. Assess cardiac, respiratory, and neurological status.

4. Be alert for possible head/neck injury.

-Secure airway.

-Use caution if C-spine injury is suspected.

-Administer O2 and assist ventilations as needed.

-Consider CPAP per the protocol.

Stabilize neck with C-collar, and backboard if spinal injury is suspected.

Treat dysrhythmias per appropriate protocol.

-Start an IV, TKO or as directed.

-Place on cardiac monitor.

Consider administration of Sodium Bicarbonate 1 meq/kg if patient has had prolonged submersion.

Transport monitoring vital signs and cardiac rhythm.

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## NEONATAL RESUSCITATION

<http://easycalculation.com/medical/Apgar-score.php>

### APGAR SCORING

#### Assessment

1. Assess airway, breathing, and circulation. The need for resuscitation can be guided by heart rate, respiratory rate and effort, tone, and color.
2. Determine APGAR score.
3. Look for meconium stained fluid.
4. Resuscitation of the neonate rarely requires advanced life support. It should proceed in a stepwise fashion with continuing reassessment to avoid overly aggressive intervention.
5. Use of Broselow Tape is recommended for all pediatric dosing.

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## NEONATAL RESUSCITATION (CONT)

\*Keep neonate dry and warm.

-Hypothermia from being wet can stress the neonate.

If the heart rate is < 60 beats/minute begin chest compressions at a rate of 100-120/minute.

-Consider Narcan 0.1 mg/kg IV, IO, IM.

-Consider Dextrose administration (refer to hypoglycemia protocol).

Position on back with head in a neutral position.

If amniotic fluid is stained with thick meconium or infant is depressed with thin meconium

-

Suction mouth and nose with bulb syringe.

-Suction mouth first.

-Do not suction for more than 10 seconds at a time.

-Suction while removing catheter only.

-Suctioning can cause reflex bradycardia if catheter is passed into hypopharynx or below.

Initially administer O2 by facemask.

If no meconium present

-Intubate immediately.

-Apply suction with meconium adapter directly to the ET tube.

-Repeat as needed until clear.

-If neonate deteriorates, stop suctioning and move on to resuscitation.

-Keep neonate warm and dry.

-If necessary tactile stimulation may be used to stimulate adequate breathing.

-If necessary, ventilate with BVM at 20 to 30 breaths/min.

-It may be necessary to disable the pop-off valve of the BVM for the initial ventilations.

-Take care not to over-inflate the lungs.

If evidence of respiratory distress despite adequate warming, drying, and oxygen treatment; intubate and gently ventilate using BVM and 100% O2.

Consider Epinephrine 1:10,000 concentration, 0.01 mg/kg IV, IO  
ET 1:1000 0.1 mg/kg



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## POISONING

### Assessment

1. Check vital signs. Monitor SpO2 and EtCO2
2. Obtain brief history. Bring any containers, package inserts, etc. of suspected ingestion. Small amounts of ingested alcohol may cause unconsciousness and/or hypoglycemia in children.
3. Obtain 12 lead EKG as soon as possible.
4. Assess level of consciousness, cardiac, and pulmonary systems. Monitor for seizure and/or respiratory distress.
5. Consider contacting Poison Control 1-800-222-1222 or medical control for information.

Alert patients with oral ingestion:

Transport monitoring vitals and level of consciousness.

Patients With Decreased Level of Consciousness:

-Do a glucose check.

-Consider administration of Dextrose per hypoglycemia protocol

-Secure airway, suction as needed.

-Administer O2 as appropriate.

-Establish an IV at a TKO rate.

-Consider administration of Narcan 0.4-2mg SIVP, titrate to respiratory effort, monitor EtCO2.

-If IV access is unavailable or patient's condition is determined to be life threatening, consider immediate administration of IN Narcan. See intranasal procedure

-Narcan IV is preferred but also IM, IO, ET or IN.

Monitor cardiac rhythm.

Transport in lateral position.

For tricyclic antidepressant overdoses with wide complex tachycardias, consider

intubation. 12 Lead EKG,

Consider administration of

Sodium Bicarbonate, 1

meq/kg IV (max 50 meq).

Contact Medical Control

prior to administration.

\*\*See commonly prescribed

TCA's\*\*

For calcium channel blocker

overdoses with hypotension,

consider administration of Calcium

Chloride 10% solution, 500 to 1000

mg or a dose of 5-10ML SIVP

Precipitates with NaHCO3

\*\*See list of commonly prescribed

CCB's\*\*

For Organophosphate or

Carbonate poisoning, administer:

Atropine 2 mg SIVP If no

tachycardia or pupil dilation,

give a second dose of 2mg

SIVP.

Seizure Activity:

Versed per seizure protocol

For a known

benzodiazepine

overdose: monitor vitals,

IV, and secure airway.

06/17/2026

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## SEIZURES

### Assessment

1. Check vital signs.
2. Obtain a brief history including an exact description of the incident from observers.

Also obtain history of medication use, presence/absence of alcohol, substance abuse, recent illness, or trauma.

3. Briefly assess cardiopulmonary and neurological system's noting any seizure activity you observe.

Single Seizure with Typical

Postictal Phase

Repetitive Seizures

-Ensure airway.

Nasopharyngeal airway may be helpful in this situation.

Administer O2 as appropriate.

Suction as needed.

Consider starting an IV.

Transport monitoring vital signs.

Secure the airway, using the appropriate airway adjunct.

-Administer O2.

-Place on cardiac monitor.

-Start an IV, TKO.

Do a glucose check.

Consider: Versed

- Children: 0.2 mg/kg Not to exceed 10mg. Give half the dose in each nostril.

- Adults: 5-10 mg SIVP over 1 minute. IN 5mg in each nare

- Additional Doses: May be given every 3-5 minutes until seizure activity stops.

-

\*Watch for respiratory depression.

Consider administration of

Dextrose, if indicated per

hypoglycemia protocol.

Transport monitoring vital signs and cardiac rhythm.

Pro

Eclampsia

Secure airway.

-Administer O2 as appropriate.

-Place on cardiac monitor.

-Start an IV, TKO

Consider Versed: 5-10 mg slow IV.

- Adults: 5 mg slow IV over 1 minute.

- Additional Doses:

May be given every 3-5 minutes until seizure activity stops.

Administer Magnesium

Sulfate 2-4g SIVP over 2-4 minutes.

Transport monitoring vital signs and cardiac rhythm.

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## SYNCOPE

Many cases of syncope are vasovagal in nature. However other causes to be considered include dysrhythmias, poisonings, trauma, or shock of various etiologies.

### Assessment

1. Check vital signs, consider Orthostatic vitals
2. Obtain brief history including onset, duration, warning symptoms (dizziness, nausea, lightheadedness), and precipitating factors (sudden change of position).
3. Perform cardiac and neurological assessments.

Stable Vital Signs With A

Normal LOC

Unstable Vital Signs With An

Altered LOC

Transport in position of comfort.

Monitor vital signs.

Monitor ECG and obtain 12/15

lead.

Consider an IV TKO, especially in

patients > 40 years of age.

Secure the airway.

Administer O2 as appropriate.

Start an IV TKO.

Monitor ECG and obtain 12/15

lead.

Proceed to appropriate protocol for

further treatment. Examples:

-Poisoning

-Hyperthermia

-Hypothermia

-Hypoglycemia

-Dysrhythmias

-Chest pain

## GENERAL TRAUMA PROTOCOL

### Assessment

1. Assess adequacy of airway and breathing with simultaneous cervical spine immobilization.
2. Assess cardiopulmonary system with attention to adequacy of perfusion (mental status, skin moisture and temperature, capillary refill, and character of pulses).
3. Use of EtCO<sub>2</sub> is recommended to determine and monitor compensation/perfusion.
4. Assess neurological function (LOC, pupils, motor function, sensation)
5. Obtain brief history of incident and mechanism of injury. Determine transport status.

### Anatomic Criteria

- Pain, deformity or tenderness to neck or back.
- Neurological Deficit.
- Painfully distracting injury.
- Pain produced on unassisted rotation of the head 45 degrees in either direction.

1. Oral/nasal tracheal intubation may be attempted if unable to adequately ventilate the patient with a BVM due to severe facial trauma, or excessive blood or secretions. Maintain in-line C-spine immobilization during intubation attempts.
2. In adults, nasotracheal intubation may be attempted if not contraindicated.
3. Cricothyrotomy may be attempted only if unable to otherwise open the airway. C-spine immobilization must be maintained during attempts.

### Physiological Criteria

- Blood pressure <90 mm/hg or absence of peripheral pulses.
- Respiratory rate >29 or <10.
- GCS < 14.
- 

### Mechanism of Injury

- Rollover
- Occupant ejection.
- MVC with intrusion greater than 1 foot.
- Fall greater than 3 feet.
- Death of same car occupant.
- Motorcycle / ATV speed >20mph.
- Prolonged extrication time >20 minutes.
- Pedestrian struck at speed >20 mph.
- Blunt Trauma.
- Hanging.
- Unrestrained with significant MOI.
- Axial Loading / Compression injury.

Secure the airway and assure adequate ventilation.

- Stabilize the C-spine with manual in-line immobilization.
  - A modified jaw thrust is recommended to open the airway.
  - Consider the use of advanced airway adjuncts, and administer appropriate O<sub>2</sub> concentration.
- For a severe extremity laceration or amputation with massive hemorrhage consider the early use of a tourniquet.

Transport of the unstable trauma patient should not be delayed by intubation attempts, unless the patient cannot be adequately ventilated with a BVM.

### Damage Control Resuscitation

- Control external bleeding sterile dressings and direct pressure or tourniquet.
- Support Airway and Breathing
- Occlusive dressing on open wounds neck to navel

Transport of the unstable patient should not be delayed to initiate IV therapy.

If Cardiovascularly Unstable:

JUDICIOUSLY administer warmed IV fluids LESS IS MORE in hemorrhagic patients (THESE PATIENTS NEED BLOOD)  
Administer 250ml bolus reassess can administer additional 250ml.

If the patient remains in shock following the first two fluid boluses, subsequent 20 ml/kg boluses may be

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## GENERAL TRAUMA (CONT)

### Spinal Immobilization

#### Protocol Consideration:

Patients with penetrating injuries should NOT be immobilized unless a focal neurologic deficit is

noted on physical examination.

In the stable patient, spinal immobilization as indicated by mechanism and exam, using the appropriate devices and techniques.

In the unstable patient, appropriate rapid extrication techniques should be utilized.

USE CAUTION, look for occult fracture in >65 or <4 y/o or a HX of osteoporosis, bone or vertebral disease.

In a pregnant patient, if the uterus is palpable above the umbilicus, place patient left lateral recumbent to

prevent compression of the vena cava and maternal hypotensive syndrome.

#### Procedure:

1. Stabilize head and neck in neutral position unless movement causes pain, deformity, or resistance. If so,

immobilize the head and neck in the position found.

2. If the patient is ambulatory, apply cervical collar and assist / position them directly on the ambulance

cot.

3. Patients that are stable, alert and without neurological deficits may be allowed to self-extricate with

assistance to the ambulance cot after placement of a cervical collar.

4. If a Long Spine Board or Scoop Stretcher is used for extrication or patient movement, the patient should

be taken off (using proper techniques) the long spine board or scoop stretcher and placed directly

on the ambulance cot.

5. Limit movement of the spine during the procedures.

Spinal immobilization for athletes and motorcyclists:

The helmet and/or shoulder pads should be removed as a unit if:

Immobilization of the helmet does not also immobilize the head.

The airway cannot be controlled or ventilation provided even after removal of the facemask.

The helmet or pads prevents immobilization for transportation in an appropriate position.

#### Special Considerations:

Older adults: low impact mechanisms (e.g., ground-level falls) might result in severe injury, anticoagulants/bleeding disorders.

Patients with head injury are at high risk for rapid deterioration

#### Burns:

Without other trauma mechanism: triage to burn facility

With trauma mechanism: triage to trauma center

Pregnancy >20 weeks

EMS provider judgment

**EMERGENT Transport!!!**

-Monitor vital signs and continually reassess patient status and interventions.

-Transport to the closest appropriate trauma center.

-Trauma codes should be taken to the nearest Level I or II trauma center, whether the trauma center is on diversion

status or not. This applies only to trauma codes.

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## ABDOMINAL TRAUMA

### Assessment

1. Assess per the general trauma protocol.

2. Briefly assess chest, abdomen, and genitalia.

Treat as per the general trauma protocol.

-If evisceration is present cover exposed viscera with a sterile saline-soaked covering and or occlusive dressing.

-Do not attempt to replace exposed viscera.

-If pelvic fracture is suspected, place patient on a long spine board, improvise padding, belt device or PASG for stabilization.

Transport to the appropriate trauma facility.

Conserve body temperature KEEP PATIENT

WARM

05/23/2025

## BURNS

<http://easycalculation.com/medical/burn-fluids.php>

### Assessment

1. Stop the burning process. Remove the patient from the source.
2. Obtain brief history noting time exposed, contact with steam, smoke or toxic gas.
3. Assess burn noting depth and extent (rule of nines). Assess cardiac and respiratory status noting any sooty sputum, singed nasal hair, or hoarseness.
4. Consider intubation or cric insertion if laryngeal swelling is present.
5. Be alert for associated injuries.

-Secure airway IMMEDIATELY.

-Administer O2, humidified if possible.

-Assist ventilations as needed.

-Start an IV. Consider 20 ml/kg fluid bolus initially (maximum of 1000 ml).

-Reassess breath sounds, capillary refill, pulse rate, and blood pressure.

\*If needed, administer a second bolus of 20 ml/kg (maximum of 1000 ml) to a total bolus of 40 ml/kg (maximum of 2000 ml).

Contact medical control for transport routing to the appropriate facility. According to MARCER,

EMS providers should now consider routing Adult burn patients directly to a burn center, regardless of the patients' age. MARCER defines a direct route patient to a burn center as one

with more than 20% second degree or greater than 5% third degree burns shall be transported to

KU Burn Unit. Pediatric burn patients shall be routed to Childrens Mercy Hospital.

Monitor ECG, EtCO2 and

CO-Oximetry

-If >20% of burns are second or third degree, cover with dry sterile burn sheet.

-Leave blisters intact.

First or second degree burns with <20% involvement, cool the area with saline soaks.

Chemical burns: remove contaminated clothing. Refer to MSDS sheet, Haz Mat ERG, and medical control for further instructions.

-Consider Morphine 2 mg increments slow IV or IM up to a total dose of 10mg or

-Consider Fentanyl 50-100 mcg slow IV, IM or IN.

Contact medical control for further orders.

Adult Pediatric

06/15/2026

## CHEST TRAUMA

### Assessment

1. Assess per general trauma protocol.
2. Perform brief chest exam. Specifically check for open wounds; flail segments, tracheal deviation, unequal breath sounds, subcutaneous emphysema, and adequacy of ventilation.

Treat as per general trauma protocol.

Splint flail segments externally with direct pressure, bulky dressings, or place patient on injured side.

For an open chest wound, cover with a commercial chest seal or occlusive dressing secured on three sides to act as a flap valve.

Transport to appropriate trauma facility.

Conserve body temperature KEEP PATIENT

### WARM

Hemo/Pneumothorax can occur with or without penetrating chest injury. If suspected, treat per plural decompression protocol

Monitor cardiac rhythm and treat dysrhythmias per appropriate protocol.

Use of nitrous oxide is contraindicated in chest trauma.



06/15/2026

## HEAD TRAUMA

### Assessment

1. Assess per general trauma protocol.
2. Obtain brief history noting mechanism of injury and level of consciousness since the injury.
3. Check for associated injuries. Assume spinal injury in all patients with a head injury.
4. High flow oxygen is not recommended, maintain saturation levels 95% or above.

Treat as per general trauma protocol.

-Immobilize C-spine.

-Administer O2 low concentration

-Start an IV, TKO or as directed.

-Assist ventilations with bag valve mask if needed.

-If intubation is indicated: consider only oral intubation.

-You must maintain manual inline stabilization with all intubation attempts.

Transport to appropriate trauma facility.

Conserve body temperature KEEP PATIENT WARM

If vitals are stable, consider elevating head 30 degrees.

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## ORTHOPEDIC EXTREMITY TRAUMA

### Assessment

1. Assess per general trauma protocol.
2. Briefly assess the injured extremity noting neurovascular adequacy (color, temperature, deformity, open wounds, and distal sensation and movement).  
Treat as per general trauma protocol.

Administer O2 as needed.

-Immobilize patient with appropriate splinting device.

-The splint should include the joint above and below the fracture.

-Severely angulated fractures may be straightened by gentle continuous traction if necessary for extrication, immobilization, or significant neurovascular compromise.

Transport to an appropriate trauma facility.

Conserve body temperature KEEP PATIENT WARM

Dress open wounds, for severe hemorrhage or amputation consider the early use of a tourniquet.

Assess neurovascular status before and after splinting.

Consider Nitrous Oxide, self administered by patient.

Isolated extremity trauma only.

Consider Morphine: 2 mg increments up to a total of 10 mg, slow IV or IM.

Or

Consider Fentanyl: 50-100 mcg slow IV, IM or IN.

Contact medical control for additional orders for Morphine or Fentanyl. (example: extended scene time)

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## SPINAL TRAUMA

### Assessment

1. Assess per general trauma protocol.
2. Perform a brief neurological exam to include motor and sensory status.

Treat as per general trauma protocol.

-Using manual in-line stabilization, apply appropriately sized C-collar.

-Use appropriate immobilization devices (long board, KED).

Establish IV access.

Transport to appropriate Trauma facility.

Conserve body temperature KEEP PATIENT WARM

Spinal (neurogenic) shock -Prevent heat loss.

-Administer 20 ml/kg (maximum of 1000 ml) bolus initially and assess the response to the bolus.

-Bolus may be repeated one time to a total of 40 ml/kg (maximum of 2000 ml).

Reassess

lung sounds  
repeatedly

08/05/2026

## ANALGESIC MEDICATION ADMINISTRATION

### Indications

The relief of moderate to severe acute pain in patients with:

- A. An extremity injury
- B. A shoulder or hip injury
- C. Back and Abdominal pain (Non-Traumatic)
- D. Burns without airway or respiratory compromise
- E. Cardiac Chest Pain

### Contraindications

- 1. Trauma
  - A. Head injury
  - B. Suspected spinal injury
  - C. Thoraco-abdominal trauma
- 2. Systems
  - A. Respiratory distress or compromise (asthma or COPD)
  - B. Cardiac dysrhythmia present
  - C. Altered mental status
- D. Pregnancy
- 3. Drugs
  - A. Sensitivity or allergy to opiates
- 4. Hypotension and suspected shock
- 5. Age less than 12 months.

-Determine that the patient meets the above criteria.

-Establish an IV at a TKO rate.

-Place patient on cardiac monitor.

-Administer oxygen.

-Place patient on pulse oximeter.

-Consider Morphine in 2 mg increments slow IV, IM up to a total dose of 10 mg for extremity, shoulder, hip, burns & Chest pain.

-

2-4mg for non-traumatic abdominal and back pain.

- For pediatric patients: 0.05 mg/kg (maximum of 2 mg).

Contact medical control for additional doses of morphine.

- Consider Fentanyl 50-100 mcg slow IV, IM or IN for extremity, shoulder, hip, & burns.

25-50 mcg for non-traumatic abdominal and back pain.

-For pediatric patients: 1-2 mcg/kg slow IV, IM or IN

-Contact medical control for additional doses.

-Contact medical control for use with burns.

Narcan, 0.4 -2 mg SIVP titrate to respiratory effort, may be used to reverse respiratory depression and hypotensive effects as needed.

Monitor EtCO<sub>2</sub>

## BIPAP

### Indications

1. Short-term management of acute respiratory distress or failure in an awake cooperative patient.
2. Near drowning patients that are awake and cooperative.

### Contraindications

1. Need for immediate intubation.
2. Unstable respiratory drive (inability to maintain their own airway.)
3. Ventilatory failure
4. Gastric distention
5. Claustrophobia

### Precautions

1. Requires patient cooperation. (The major complication is the inability to tolerate the mask; in which case the mask should be removed and an alternate airway should be used.)
2. If patient complains of nausea, remove mask. The mask may be held in place manually. (Vomiting with the mask in place virtually assures aspiration.)
3. Adequate supply of oxygen is required.

Inform patient of the procedure.

Settings for BiLevel (BL) should be

FIO2 set at 100%

IPAP 12 cm/Hg

EPAP 6cm/Hg

LC compensation on

Turn on the oxygen, hold the mask firmly against the patient and instruct the patient to take slow deep breaths, relaxing and allowing the machine to help.

-After the patient has tolerated the mask, the straps may be attached.

-Monitor the patient for comfort, anxiety, and nausea.

The mask must be unstrapped if any nausea develops, but may be held in place manually by the patient with coaching.

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## CPAP

### Indications

- 1 Short-term management of acute respiratory distress or failure in an awake cooperative patient.
- 2 Near drowning patients that are awake and cooperative.

### Contraindications

- 1 Need for immediate intubation.
- 2 Unstable respiratory drive (inability to maintain their own airway.)
- 3 Ventilatory failure
- 4 Gastric distention
- 5 Claustrophobia

### Precautions

1. Requires patient cooperation. (The major complication is the inability to tolerate the mask; in which case the mask should be removed and an alternate airway should be used.)
2. If patient complains of nausea, remove mask. The mask may be held in place manually. (Vomiting with the mask in place virtually assures aspiration.)
3. Adequate supply of oxygen is required.

Inform patient of the procedure.

Turn on the oxygen, hold the mask firmly against the patient and instruct the patient to take slow deep breaths, relaxing and allowing the machine to help.

-After the patient has tolerated the mask, the straps may be attached.

-Monitor the patient for comfort, anxiety, and nausea.

The mask must be unstrapped if any nausea develops, but may be held in place manually by the patient with coaching.

## Double Sequential External Defibrillation

### Indications

Refractory Ventricular Fibrillation or Pulseless Ventricular Tachycardia resistant to predefibrillation attempts and pharmacological efforts.

After two attempts of defibrillation and the administration of an antiarrhythmic, consider double sequential.

Place second set of pads on anterior and posterior regions from second monitor.

Simultaneously charge both monitors at 200 Joules.

Defibrillate the patient with both monitors simultaneously.

ROSC Achieved?

See Return of Spontaneous Circulation Protocol.

### Contraindications

Pediatric cardiac arrest

ROSC Not Achieved?

Return to needed cardiac arrest protocol.

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## EZ-IO INSERTION PROCEDURE

### Indications

1. For use whenever traditional IV access is not available, or after three unsuccessful IV attempts.
2. For use with patients between 3 -39 kg when peripheral IV access cannot be established in 2 attempts or 90 seconds.

### Contraindications

1. Fracture of tibia, humerus or femur.
  2. Previous orthopedic procedures (knee placements)
  3. Any infection over the insertion site.
  4. Inability to locate the three anatomical landmarks.
  5. Excessive tissue over the insertion site.
  6. Needles are one time use only.
  7. Never attempt a second insertion in the same leg.
- Determine if use is indicated.

- Ensure there are no contraindications present.
- Locate proper insertion site (about 1 finger width medial to the (humeral tibial) tuberosity on the flat aspect of the bone).
- Cleanse the insertion site.
- Stabilize (leg, upper arm) with one hand.
- Insert at a 90 angle to the surface of the bone.
- Begin insertion.

Prepare the driver to the needle set.

- Stop insertion when the needle flange touches the skin or a sudden decrease in resistance is felt.
- Remove the driver and stylet.
- Confirm proper placement by aspirating a small amount of bone marrow with a syringe.
- Attach the extension and flush with 10 ml of fluid.
- Administer the medication or begin the fluid infusion.
- Apply the identification wristband and apply a dressing to the insertion site.

Consider

giving

Lidocaine

2%: 25 mg

slow IO

before

flushing the

IO line for the

conscious

patient,

unless there

is a known

allergy.



7/25/2023

#### EZ-IO INSERTION CONTINUED

##### SHOULDER INSERTION

-In the event the lower extremities are unavailable for IO access, the head of the humerus may be used.

-Patients must be above the age of 8 and/or >40 kg.

-Patients should be in a supine position.

-Expose the shoulder and adduct the humerus.

-Place the patient's hand on the umbilicus and ensure the elbow remains in contact with the stretcher or ground.

-Palpate the humeral shaft with the thumb progressing superiorly toward the humeral head, palpating for the greater tubercle.

-As you near the shoulder you will feel a protrusion. This is the base of the greater tubercle, your insertion site.

-Confirm the insertion site by palpating the outer margins of the tubercle. This should be slightly anterior to the arms midline.

-Once the landmark is identified, follow the insertion procedure.

##### PEDIATRIC INSERTION

The tibial tuberosity may be difficult or impossible to palpate on very young patients. This insertion technique shall be used on patients 3-12 kg.

Insertion site is located 2 finger widths below the patella, then medial along the flat aspect of the tibia.

On patients 13-39 kg where the tibial tuberosity is palpable, the landmark is one finger width below the tibial tuberosity along the flat aspect of the tibia.

-Gently guide the needle in at a 90 angle.

-Do not push.

-Allow the driver to do the work. Stop when you feel the pop.

-Excessive force can cause a widening of the needle entry point increasing the risk of extravasation.

-Flush the site with 5 ml of fluid.

-In patient's <40 kg, shoulder insertion is not allowed.

05/23/2025

## IMPLANTABLE CARDIOVERTER - DEFIBRILLATOR (ICD)

### DE-ACTIVATION

#### Indications

ICD malfunctioning causing inappropriate shocks to rhythms other than VF/VT.

#### Contraindications

1. ICD functioning properly and administering appropriate shocks to VF/VT.
2. EMS not witnessing the shocks by the ICD.
3. Patient not on cardiac monitor.

#### ICD Functioning

##### Appropriately With

##### Shocks To VF/VT

##### ICD Malfunctioning

Apply fast patches.

#### ICD Functioning

##### Appropriately With No

##### Shocks Observed

If patient is unstable

-Perform immediate  
synchronized

cardioversion: 100-150-200 J

-Establish IV access &

give sedation; do not  
delay cardioversion.

If patient is stable

Refer to wide complex

tachydysrhythmia

protocol.

If pulseless arrest

develops, see

pulseless arrest

algorithm.

Apply magnet over

ICD to deactivate.

#### Assessment

1. Assess and support ABC's.
2. Determine cardiac rhythm and monitor continuously.
3. Assess cardiac and respiratory systems and level of consciousness.

Transport monitoring

vital signs.

General Treatment -Monitor blood pressure, oximetry, and administer oxygen as appropriate.

-Start an IV, TKO or as directed.

Secure magnet over

ICD with tape to

maintain deactivated

status.

06/15/2026

## INTRANASAL ADMINISTRATION PROCEDURE

### Indications

For the treatment of persistent seizure activity, narcotic drug overdose, Pain relief and sedation when IV access is unobtainable, or a life threatening condition exists.

### Contraindications

1. When IV access is obtainable.
2. If unable to clear blood or mucus from nostrils.
3. Sensitivity to benzodiazepines.
4. 0.5 cc per nostril is considered well tolerated in adult and pediatric patients

-Inspect nostrils for mucus, blood, or other obstructions that might inhibit absorption.

-Obstructions must be suctioned.

-For persistent seizures, Use 5mg/ml Versed

For Adults; administer, 5mg in each nare.

-For pediatrics: administer 0.2mg/kg dose not to exceed 10 mg.

-For narcotic drug overdose or coma of unknown etiology, consider a one time 4mg Narcan spray or Narcan 0.4 -2 mg titrated to respiratory effort.

### Pediatric 0.1 mg/kg

-Administer one half the volume in each nostril.

-Must use a 1 mg/ml solution.

-Attach atomizer to syringe.

-Administer one half the volume in each nostril.

Fentanyl 50-100 mcg slow IV, IM or IN.

-For pediatric patients: 1-2 mcg/kg

05/23/2025

## NEEDLE / SURGICAL CRICOTHYROTOMY AIRWAY PROCEDURE

### Indications

1. Upper airway obstruction that is not relieved by BLS measures or direct laryngoscopy with removal of object by magill forceps.
2. Severe facial trauma, which makes obtaining an airway by less invasive means impossible.

### Contraindications

1. Neck swelling that obscures landmarks.
2. Not a prophylactic measure. Must actually need an airway and ventilation now.

### Precautions

1. Contact medical control after procedure.

### Technique

1. Locate the cricothyroid membrane between the thyroid and cricoid cartilage.
2. Cleanse the area with betadine or alcohol as time permits.
3. Using the scalpel provided in the kit, make a 2 cm vertical incision over the cricothyroid membrane.
4. Separate the incision and locate the cricothyroid membrane.
5. Using the scalpel, make an incision through the cricothyroid membrane.
6. Insert the dilator and tube into the trachea.
7. Remove the dilator and inflate the cuff of the tube.
8. Control any bleeding and apply sterile dressing around area of the incision.

### Complications

1. Placement at the wrong level can cause laryngeal or tracheal damage.
2. Bleeding
3. Massive subcutaneous and/or mediastinal emphysema
4. Pneumothorax

05/23/2025

**PUSH DOSE EPINEPHRINE 1:100,000**

Indications: Used to treat unremitting hypotension after appropriate fluid resuscitation in post-cardiac arrest, cardiogenic shock, or other medical etiology, in bradycardia if poor perfusion persists secondary to transcutaneous pacing and appropriate fluid resuscitation.

Contraindications: Cardiac Arrest, Trauma etiology, Pediatric patients or Known hypersensitivity

Precautions: Use caution in patients with suspected myocardial infarction or history of coronary artery disease. In patients taking digitalis, epinephrine may exacerbate ventricular ectopy.

Doses: 10-20mcg every 2-3 minutes if SBP < 90mmHg Route: IV/IO Protocol Pages: Bradycardia & Hypotension

Technique: Push Dose Epinephrine

1. Draw up 9mL of normal saline into an empty 10mL syringe
  - a. A 10mL flush may also be used after 1mL is wasted
2. Into the same syringe, draw up 1mL of epinephrine from the cardiac amp (1:10,000)
3. Syringe now contains 10mLs of Epinephrine at (1:100,000) or 10 mcg/mL
4. Dose is 1-2 mL every 2-3 minutes as needed

\* Onset- 1 minute

\* Duration- 5-10minutes

\* Dose-0.5ml-2ml every 2-5 minutes (5-20 mcg)

Mark or label syringe with the mix of 1:100,000 as soon as it's made up to prevent medication administration error.

05/23/2025

## PLEURAL DECOMPRESSION PLACE OCCLUSIVE DRESSING

### Indications

Tension pneumothorax should be considered when the following signs/symptoms are present:

1. Progressive severe respiratory distress.
2. Decreased or absent breath sounds on affected side.
3. Jugular vein distension.
4. Tracheal deviation away from the affected side.
5. Tympany to percussion on the affected side.
6. Cardiac compromise.

### Contraindications

1. Not sure of diagnosis.
2. Clinical signs suggestive of diagnosis, but patient is not in shock and/or respiratory distress.

### Precautions

1. Contact medical control after procedure.

### Technique

1. Locate the second and third intercostal space in the mid-clavicular line on the affected side of the chest, or the fifth and sixth intercostal space in the anterior-axillary line.
2. Cleanse with betadine or alcohol preps as time allows.
3. Insert a 14-gauge catheter on the superior plane of the rib. Withdraw the needle and attach a flutter valve to the catheter. Secure catheter to the chest.
4. Re-evaluate respiratory status and vital signs.

### Complications

1. Creation of a pneumothorax if one is not present.
2. Damage to lung or viscera.
3. Bleeding (intercostal vessels are below each rib).
4. Infection

05/23/2025

## CAPNOGRAPHY

### Clinical Indications:

Capnography shall be used when available with the use of all invasive airway procedures including but not limited to endotracheal intubation, nasotracheal intubation, cricothyrotomy, or Blind Insertion Airway Devices (BIAD). Capnography shall also be used: In accordance with current ACLS recommendations for determining effectiveness of resuscitation efforts (10 mm/Hg or above), on all patients in respiratory distress, and all patients undergoing sedation.

### Procedure:

1. Attach capnography sensors.
2. Note CO<sub>2</sub> level and waveform changes. These will be documented on each respiratory failure, cardiac arrest, or severe respiratory distress patient.
3. The capnometer shall remain in place with the airway and be monitored throughout the prehospital care and transport.
4. Any loss of CO<sub>2</sub> detection or waveform indicates an airway problem or device malfunction and should be documented.
5. The capnogram should be monitored as procedures are performed to verify or correct the airway problem.
6. Document the procedure and results on/with the Patient Care Report (PCR).

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## SEDATION

Indications For patients about to undergo a painful procedure (cardioversion, conscious endotracheal intubation)

### Contraindications

1. Known sensitivity or allergy to benzodiazepines
2. Shock

- Administer oxygen as appropriate.
- Establish an IV at a TKO rate.
- Place patient on pulse oximeter and obtain baseline saturation.
- Place patient on cardiac monitor.
- If IV access is unavailable, consider intranasal administration of Versed.

For adults: initial dose 2-4 mg IN.

-In adults you may repeat 2 mg increments up to a total dose of 10 mg.

### intranasal procedure

- Observe respirations, O2 saturation, and blood pressure.
- Administer additional O2 and assist ventilations as needed.
- Administer Versed initial dose 2-4 mg slow IV.
- In adults you may repeat 2 mg increments up to a total dose of 10 mg.



05/23/2025

## SPINAL CLEARANCE

### Indications:

To allow Paramedics to perform field clearance from full spinal restrictions.

### Contraindications

1. Any patient less than 12 years of age
2. Patients with chronic neck or back pain
3. Patients exhibiting signs of shock
4. Signs/symptoms of alcohol or drug intoxication
5. Altered Mental Status
6. Distracting Injuries

-Full spinal precautions must take place if yes was answered to any of the above questions.

\*Maintain a high index of suspicion with older/elderly patients, as occult fractures are possible with benign mechanisms of injury or trauma.

-If the patient meets the criteria to withhold spinal precautions, EMS providers may still elect to provide spinal precautions.

### Examples of distracting

#### injuries:

Extremity fractures, rib fractures, pelvic fractures, abdominal pain from trauma related injury, and other significant injuries.

### Assessment

All patients will be assessed by the following criteria. Only those of whom answer no to all requirements, may be cleared.

1. Does the patient have a GCS less than 15?
2. Does the patient complain of neck or back pain?
3. Is there tenderness, swelling, or deformity noted when the complete spine is palpated? You must visualize and palpate the entire spine: neck to lower lumbar.
4. Is there a distracting injury or distracting pain?
5. Are there signs/symptoms of alcohol or drug intoxication present?

### Examples of

#### signs/symptoms of

#### intoxication:

GCS <15, slurred speech, dilated/constricted pupils, flushed skin, unsteady gait, irregular behavior, presence of paraphernalia or admitted use.

05/23/2025

## VITAL SIGNS

### INDICATIONS:

1. A minimum of two sets of vitals will be obtained on every transported patient. This will include Pulse, Respirations, BP, Spo2 and alert and oriented status.
2. A minimum of one set of vitals will be taken on refusals.

### Technique:

1. The first set of vitals a manual blood pressure and pulse will be taken on the following calls: chest pain, stroke, overdose, syncope, trauma, hypo/hypertension, anaphylaxis and any unconscious or critical patient.
2. A manual BP will be taken if the automated device is more than 30 points different than first set obtained (systolic or diastolic) in either direction.
3. A full set of vitals will be taken after each medication administration.
4. Capillary refill and/or an automated blood pressure device is expectable for pediatric patients.

## ORTHOSTATIC VITAL SIGNS

### Indications:

1. Syncope or Near Syncope (ie. dizzy, faint)

### TECHNIQUE:

Patient in supine position for minimum of three minutes before taking lying blood pressure and pulse, then stand patient and immediately repeat blood pressure and pulse. If patient unable to stand, sit him upright with legs dangling over cot.

### SPECIAL NOTES AND EVALUATION:

1. Orthostatic changes are not valid if:
  - a. Taken initially upright, then supine.
  - b. Taken upright with only head of cot elevated and legs not dangling.
  - c. If only pulse or blood pressure is checked by itself. Must have both blood pressure and pulses in both positions.
2. Significant orthostatic changes include:
  - a. Pulse increase greater than 20 beats per minute when moved upright.
  - b. Pressure (diastolic most significant) drop greater than 20 MM Hg when upright.
  - c. Pallor, diaphoresis, or faintness when upright.
3. Significant changes could indicate volume depletion (dehydration or blood loss), treat the symptoms, follow DEHYDRATION protocol if indicated.
4. Considerations:
  - a. Treat the symptoms.
  - b. Medications can often cause changes, usually indicated by pressure change but no pulse change.
  - c. May be due to slowing of compensatory mechanisms in the elderly.

05/23/2025

## KING AIRWAY LT-D

### Indications

1. Patient who is apneic or unconscious and spontaneously breathing without an intact gag reflex, requiring airway management:
2. Two failed intubation attempts and one failed tube introducer attempt (see difficult airway algorithm).

### Contraindications

1. Known esophageal disease (varicies, alcoholism, cirrhosis etc.).
2. Known ingestion of a caustic substance.
3. Patients who are conscious or who have an intact gag reflex.

If the patient regains consciousness or begins to fight the tube

1. You may remove the King Tube.

CAUTION: When removing the King Tube there is a strong possibility of emesis.

2. You may leave the King Tube in place and sedate the patient.

### Endotracheal intubation

1. Endotracheal intubation provides a definitive airway. Every attempt following protocol should be made to secure the airway with an Endotracheal tube.
  - a. If the King Tube has been placed prior to your arrival and is functioning properly, the decision to leave it in place should be highly considered.
  - b. If deemed necessary by ALS personnel, an endotracheal tube may be inserted to take the place of the King Tube.

### Additional notes

1. Placement of the King Tube should never delay:
  - a. CPR.
  - b. Basic airway management.
  - c. Use of the defibrillation unit.
  - d. Other necessary patient care.
2. Adequate ventilation should be assessed by the use of:
  - a. Standard checks for breath sounds and visual chest rise.
  - b. Capnography.
  - c. Pulse oximetry if available.

05/23/2025

#### DIFFICULT AIRWAY ALGORITHM

\* L=Look externally (facial trauma, large incisors, beard or moustache, and large tongue)

\* E=Evaluate the 3-3-2 rule (incisor distance <3 fingerbreadths, hyoid/mental distance <3 fingerbreadths, thyroid-to-mouth distance <2 fingerbreadths)

\* M=Mallampati (Mallampati score 3)

\* O=Obstruction (presence of any condition that could cause an obstructed airway)

\* N=Neck mobility (limited neck mobility).

For each patient, the airway assessment score was compared with the Cormack-Lehane laryngoscopic view seen during intubation (1=full view, 4=glottis not visualized). At intubation, 114 patients were classified as Cormack-Lehane grade 1 (defined by the authors as easy intubation), and 42 were classified as grade 2 or higher (defined by the authors as difficult intubation). Patients in the difficult-intubation group had significantly higher LEMON scores than did those in the easyintubation group. Of the criteria used to calculate the score, only large incisors, inter-incisor distance <3 fingerbreadths, and thyroid-to-floor-ofmouth distance <2 fingerbreadths were associated significantly with difficult intubation.

#### DIFFICULT AIRWAY

Consider early use of  
Video Laryngoscope as  
well Bougie introducer.

Consider Apneic  
Oxygenation with 10  
LPM via Nasal Cannula  
for duration of  
attempts

7/25/2023

MCI TRIAGE- START/JUMP START

05/23/2025

05/23/2025

05/23/2025

ALBUTEROL (PROVENTIL, VENTOLIN)

CLASS OF DRUG

Sympathomimetic, Beta 2 selective adrenergic bronchodilator

INDICATIONS

1. Albuterol is used to treat reversible airway obstruction caused by:

- a. Wheezing associated with asthma
- b. COPD

CONTRAINDICATIONS

1. Hypersensitivity

DRUG INTERACTION

1. Beta adrenergic agents - potentiates the effects
2. MAO inhibitors - may lead to hypertensive crisis
3. Beta adrenergic blockers - decreases the effectiveness

ADMINISTRATION

Nebulizer

Adult: [2.5mg] in 3 ml of sterile NS given as inhalation therapy over 5-15 minutes, may be repeated x 2.

Pediatric: [1.25-2.5 mg] in 3 ml of sterile NS given as inhalation therapy over 5-15 minutes, may be repeated x 2.

SPECIAL NOTES

1. Most side effects are dosage related.



05/23/2025

ALBUTEROL & ATROVENT (Duoneb)

PHARMACOLOGY & ACTIONS:

A potent relatively selective beta 2 - adrenergic bronchodilator.

Has occasional beta 1 overlap with clinically significant cardiac effects.

INDICATIONS:

Bronchial asthma

Reversible bronchial spasm that occurs with chronic pulmonary disease.

CONTRAINDICATIONS:

Stop treatment if pulse increases by 20 bpm, frequent PVC's develop, any tachyarrhythmias other than sinus tachycardia appear, chest pain, apnea, nausea or vomiting, or increased shortness of breath occur.

Patients with allergies to soy or peanuts should not be administered

Atrovent

ADMINISTRATION:

Nebulizer dosage for adults & children over 1 year - 2.5 mg mixed in 3 ml of saline for a concentration of .83 mg/ml.

Infants under 1 year: Nebulizer dosage 0.03 ml/kg max dose 1ml.

Nebulization:

Oxygen flow should be set at 6 Lpm.

Patients should inhale slowly & exhale passively through nose.

SIDE EFFECTS & SPECIAL NOTES:

Patients with COPD should be monitored carefully for CO2 retention & decreased levels of consciousness.

Paradoxical bronchospasm may occur with excessive administration.

Skeletal muscle tremors

Albuterol should be used with caution in pregnancy.

Continually assess patient's respiratory rate, effort, and lung sounds.

05/23/2025

ADENOCARD (ADENOSINE)

Indications: PSVT

Contraindications: Second or third degree heart block

Known hypersensitivity to the drug

Sick-sinus syndrome or symptomatic bradycardia

Atrial fibrillation or flutter

Side Effects: Dyspnea,

Chest pressure/pain

Dizziness

Facial flushing

Arrhythmias including blocks

Precautions: Patients with asthma

Children

Dose: Initial Dose: 6 mg given as a rapid I.V. push

Second: 12mg given as a rapid I.V. push

May repeat 12 mg dose, once.

Route: IV should be established in the AC, Adenocard should be administered into the medication port closest to the heart and followed by flushing the line.

05/23/2025

# AMIODARONE

Indications: Ventricular Fibrillation

Pulseless Ventricular Tachycardia

Wide complex Tachydysrhythmias

Premature ventricular contractions

Contraindications: Known hypersensitivity to the drug or any of its components including iodine

Sinus bradycardia

Second or third degree heart block

Precautions: Exacerbation of presenting arrhythmia can occur with concurrent use of other antiarrhythmic medications (whatever antiarrhythmic you start with, stay with)

CHF

Children

Side Effects: Prolonged QT interval

Hypotension

Bradycardia

Doses: VF/Pulseless VT: initial dose is 300 mg IV. Repeat doses at 150 mg every 3-5 min up to 2.2 g

Wide complex tach: 150 mg mixed in a 100 ml bag of NS, given over 10 min (15 mg/min). May repeat at 150 mg every 10 minutes up to 2.2g.

Premature ventricular contractions (unstable):150 mg mixed in a 100 ml bag of NS, given over 10 minutes (15 mg/min). May repeat same dose every 10 min up to a 2.2 g total.

Route: IV, IO

05/23/2025

ASPIRIN

Indications: Chest pain suggestive of AMI

Contraindications: Known hypersensitivity to the drug

Patients with bleeding disorders

Children <12 with flu-like symptoms

Peptic ulcer

Precautions: GI bleeding or upset stomach

Patients with asthma

Side Effects: Heartburn

Nausea

Wheezing

GI bleeding

Reye's syndrome in children

Dose: 324 mg

Route: Oral/must be chewed

05/23/2025

## ATROPINE

Indications: Bradycardia

Organophosphate/Carbonates poisoning

Contraindications: Tachycardia/Tachydysrhythmias

Side Effects: Palpitations

Tachycardia

PVCs

HTN

Pupil dilation

Nausea/vomiting

Doses: Unstable bradycardia: 0.5 mg initially, then 1 mg IV  
every 3-5 minutes to a total dose of 3mg.

Type II Second Degree AV block and Third Degree AV block:

0.5 mg IV while awaiting pacer. May repeat to a total dose of 3 mg. If  
ineffective begin pacing.

Organophosphate/Carbonate poisoning: administer Atropine 2 mg

SIVP If no tachycardia or pupil dilation, give second dose of 2mg  
SIVP.

Pediatric Dose: Refer to Broselow Tape

Route: IV, IO, ET tube, IM (only for organophosphate poisoning)

05/23/2025

BENADRYL (DIPHENHYDRAMINE)

Indications: Allergic Reactions

Anaphylaxis with respiratory distress

Anaphylactic shock

Extrapyramidal Symptoms (EPS)

Dystonic Reactions

Precautions: Patient with asthma

Pregnancy

Lactating females

Side Effects: Drowsiness

Headache

Palpitations

Dizziness

Dilated pupils

Thickening of bronchial secretions

Dose: 25-50 mg

Pediatric Dose: 1 to 2 mg/kg

Routes: IV, IO or IM

05/23/2025

**CALCIUM CHLORIDE 10%**

Indications: Calcium channel blocker overdoses with hypotension. Examples of calcium channel blockers are Amlodipine (Norvasc), and most drugs ending in "ipine," that are used to reduce vascular resistance and arterial pressure

Contraindications: Hypercalcemia

Ventricular Fibrillation

Digitalis toxicity

Precautions: Patients on digitalis

Children

Renal disease and respiratory disease

Make sure IV is completely patent because it will cause severe tissue necrosis.

Will precipitate with  $\text{NaHCO}_3$

Side Effects: Pain or burning at IV site

Necrosis of tissue

Shortened QT

Heart block

Arrhythmias

Dose: 500 to 1000 mg (5-10 ML) over 5-10 minutes

Route: IV, IO

05/23/2025

DILTIAZEM HCL (CARDIZEM)

CLASS OF DRUG

Calcium Channel Blocker; Coronary Vasodilator, Antidysrhythmic

INDICATIONS

1. Symptomatic Atrial Fibrillation or Atrial Flutter with rapid ventricular response
2. Symptomatic Paroxysmal Supraventricular Tachycardia

CONTRAINDICATIONS

1. Sick sinus syndrome except in the presence of a functioning ventricular pacemaker.
2. Patients with second- or third degree AV block except in the presence of a functioning ventricular pacemaker.
3. Patients with severe hypotension, cardiogenic shock or severe CHF
4. Patients who have demonstrated hypersensitivity to the drug
5. Patients with atrial fibrillation or atrial flutter associated with an accessory bypass tract such as in WPW syndrome or short PR syndrome
6. Patients with ventricular tachycardia
7. Pregnancy
8. Children under the age of 15

WARNING- DILTIAZEM use in severe congestive heart failure; patients with reduced ventricular function may not be able to counteract the inotropic and chronotropic effects of diltiazem, the result being an even higher compromise of function.

DRUG INTERACTION

Additive effects in prolonging AV conduction when using beta-blockers or digitalis concomitantly with Diltiazem HCl.

ADMINISTRATION (For patients over age 70 decrease dose by 5mg)

0.25 mg/kg max 20mg SIVP over 2 minutes, continue to monitor the PT, if hypotension develops, DC medication and consider a fluid bolus. Repeat 12 lead ECG if rhythm converts. If no response after 10 minutes, consider a second dose of 0.35mg/kg max 20mg SIVP over 2 minutes. If hypotension develops DC medication and consider a fluid bolus. Repeat 12 lead ECG if the rhythm converts.

SPECIAL NOTES

\*Pediatric: Not usually used.

\*DO NOT TREAT BRADYCARDIA WITH ATROPINE, DILTIAZEM WILL WEAR OFF.

\*For patients over age 70 decrease dose by 5mg.



07/07/2026

Dextrose: D10 & Oral Glucose

Indications: Hypoglycemia (< 60 mg/dL) and Altered Mental Status (GCS < 14)

Contraindications: Hyperglycemia

Precautions:

Side Effects:

Dose:

Pediatric Dose:

Delirium tremens

Cranial/spinal hemorrhage

CHF

May cause necrosis if infiltrated into tissue

Local venous irritation

Pulmonary edema

Hyperglycemia

For adults, give D10 (250mL) 25g or equivalent IV.

Children greater than 8 years of age should receive 2 mL/kg of D10 IV

Children less than 8 years of age should receive 1 mL/kg of D10 IV

**\*\*Refer to the Broselow tape for weight estimation as needed.\*\***

15g oral glucose may be considered for awake, cooperative patients  
with a patent airway and the ability to swallow

Oral Dose:

05/23/2025

DOPAMINE

Indications: Cardiogenic shock (hypotension)

Type II second degree AV block

Third degree AV block

Contraindications: Hypersensitivity

VF

Tachydysrhythmias

Precautions: Peripheral vascular disease

Will cause tissue necrosis with extravasation of IV

Side Effects: Headache

Palpitations

Tachycardia

HTN

Ectopic beats

Wide QRS complex

Nausea/vomiting

Necrosis of tissue

Doses: Type II second degree & Third degree AV blocks: 2-10 mcg/kg/min

Hypotension of cardiac origin: 2-10 mcg/kg/min. Titrate to a  
blood pressure of 100.

Route: IV drip only

05/23/2025

EPINEPHRINE 1:1,000

Indications: Asthma

Mild anaphylaxis

Anaphylaxis with respiratory distress

Anaphylactic shock

Children with unstable bradycardias

Contraindications: Over 40 years old AND

History of CAD

Heart rate over 120

Must contact Medical Control prior to administering to  
above patients

Precautions: Pregnancy

HTN

Side Effects: Tremors

Anxiety

Palpitations

Tachycardia

HTN

Dysrhythmia

Dyspnea

Doses: Severe Asthma: 0.3 mg IM

Mild anaphylaxis/Anaphylaxis with respiratory distress: 0.3 - 0.5 mg IM

Anaphylactic shock: 0.3 - 0.5 mg max given via injection in the venous  
plexus at the base of the tongue.

Pediatric Dose: Refer to Broselow Tape

Route: IV, IO, IM, IN, ET

05/23/2025

EPINEPHRINE 1:10,000

Indications: Asystole/PEA

Persistent or recurrent VF/VT

Unstable brady dysrhythmias in children

Pediatric Anaphylaxis

Contraindications: None in pulseless arrest situations

Precautions: None in pulseless arrest situations

Doses: Pulseless Arrest: 1 mg IV/IO every 3-5 minutes or 2 mg ET

Pediatric/Neonatal Dose:

Anaphylaxis 0.01 mg/kg

Route: IM (Refer to Broselow Tape)

Cardiac 0.01 mg/kg

Route: IV, IO, and ET (Refer to Broselow Tape)

Route: IM, IV, IO, ET x 2

04/01/2026

FENTANYL

Indications: Pain from the following:

An extremity injury

A shoulder or hip injury

Back pain and Abdominal pain (non-traumatic)

Burns without airway or respiratory compromise - (Contact Med Control)

Contraindications: Known hypersensitivity

Precautions:

Side Effects:

Dose:

Pediatric

Dose: Route:

Respiratory depression

Must be given slow or it can cause muscle rigidity (stone chest or rigid chest syndrome)

Muscle rigidity

Bradycardia

Respiratory depression

Laryngospasm

50-100 mcg slow IV for extremity pain

25-50 mcg slow IV for non-traumatic back or abdominal pain

Must contact medical control for use with burns.

1 to 2 mcg/kg IV, IN

IV, IO, IN

06/15/2026

## GLUCAGON

Must transport patient after administration.

Indications: Hypoglycemia ( $< 60$  mg/dL) and Altered Mental Status (GCS  $< 14$ ) when unable to obtain IV access

Contraindications: IV/IO access

Precautions:

Side Effects:

Adult Dose:

Known sensitivity

Only effective in patients with sufficient amounts of glucose stored in the liver (glycogen). So use with caution in patients with prolonged fasting, starvation, or chronic hypoglycemia.

Secondary hypoglycemia can occur, and is counteracted by sufficient carbohydrate intake.

Pregnancy

Nausea

Vomiting

Hypotension

Hyperglycemia

1 mg

Pediatric Dose: Pediatrics greater than 25kg give 1 mg

Pediatrics less than 25kg give 0.5 mg

Route: IM

#### Ketamine

Contraindications: Known sensitivity, hypoxia, head or airway trauma

Indications: Extreme agitation/Hyper-agitated delirium syndrome

Precautions: Use with caution in intoxicated patients or patients who have already received sedating medications.

Time to onset: IM dosing 3-4 minutes

Side effects: 1. Blood pressure and pulse are frequently elevated.

2. Can result in patient dizziness and nausea/vomiting.

3. Laryngospasm and hypersalivation.

4. Emergence Phenomenon in 10-20% of adults.

5. Possible intercerebral and interocular pressure increase.

Route: IM only

Pediatric use not approved at the time.

Dose: 1. <70 kg (<154lb): 200mg IM

2. 70kg to 100kg (154lb to 220lb); 300mg IM

3. 100kg to 130kg (220lb to 286lb); 400mg IM

4. >130kg (>286lb); 500mg IM

5. If the medication is given as an adjunct (i.e. the patient already has been given sedating

medication or is intoxicated with sedating substances such as alcohol or benzodiazepines), a lower dose will be needed. 2 mg/kg is recommended as the dose in this situation.

Dissociation could occur at 3-5 mg/kg IM. May require repeat dosing based on frequent assessment. Emergence reactions are typically self limited, and can be treated with benzodiazepines. Pretreatment not proven to be effective in preventing this. Patients require

constant monitoring.

Medical control should be contacted for additional doses or subsequent treatment with any further

potentially sedating medications. This protocol is for a one time dose of Ketamine only.

05/23/2025

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05/23/2025

LIDOCAINE

Indications: VENTRICULAR FIBRILLATION or PULSELESS VENTRICULAR TACHYCARDIA

Frequent PVC's of 6 or more per minute

For anesthesia prior to fluid and drug administration through IO on a conscious patient.

Contraindications: High-degree heart blocks (2nd degree type II, 3rd degree, Bi-fascicular blocks)

Supraventricular dysrhythmias

Adams-Stokes syndrome

Wolff-Parkinson-White syndrome

Side Effects: Hypotension

Heart block

Dizziness

Convulsions

Dose: VF or Pulseless VT: 1 to 1.5 mg/kg initial dose. Repeat at 0.5 to 0.75 mg/kg, max 3 doses or 3 mg/kg. If down the ET tube, double the dose.

IO insertion: 25 mg slow IO before flushing the IO line

Pediatric Dose: Refer to Broselow Tape

Routes: IV, ET, IO



05/23/2025

MAGNESIUM SULFATE

Indications: Torsades de pointes

Eclampsia

Asthma

COPD

Contraindications: Renal disease

Side Effects:

Precautions:

Doses:

Routes:

Acute surgical abdomen

Nausea/Vomiting

Sharply lowered blood pressure

Respiratory depression

Pregnancy

Torsades: 1 to 2 g SIV/IO

Eclampsia: 2 to 4 g SIVP.

Severe Asthma: Consider 1 gm drip over 10 minutes in 100 mL

bag. Recheck vital signs, and then give the Second 1g dose  
over 10 minutes.

Severe COPD: Consider 1gm drip over 10 min in 100 ML bag. Reevaluate recheck vitals and then  
give the second 1g dose over 10  
minutes.

IV, IO

05/23/2025

METHYLPREDNISOLONE (MEDROL, SOLU-MEDROL)

CLASS OF DRUG

Glucocorticoid; anti-inflammatory; immunosuppressant

INDICATIONS

1. Asthma/COPD Exacerbation
2. Anaphylaxis

CONTRAINDICATIONS

1. Hypersensitivity

DRUG INTERACTION

1. Potential hypokalemia may increase risk of digitalis toxicity
2. May increase insulin requirement
3. Additive hypokalemia with diuretic

ADMINISTRATION

2 mg/kg IV up to 62.5 mg

SPECIAL NOTES

Use on patients with a history of diabetes.

04/01/2026

MORPHINE SULFATE

Indications: Chest pain unrelieved by NTG

Pain from orthopedic trauma or burns

Back pain & Abdominal pain (Non-Traumatic)

CHF

Contraindications: Known sensitivity

Precautions:

Side Effects:

Dose:

Pediatric Dose:

Route:

Systolic BP < 100

Pregnancy

Respiratory depression

Nausea/vomiting

Respiratory depression

Lowered BP

2 mg increments SIVP until the pain is resolved or a total dose of

10 mg is given.

2-4 mg for non-traumatic abdominal or back pain

For chest pain: 4 mg can be given initially, followed by 2 mg

increments

Must contact medical control for use with burns per Med

Director.

0.05 mg/kg slow IV/IN to a maximum dose of 2 mg.

IV, IM

06/17/2026

**NARCAN (NALOXONE)**

Indications: Narcotic overdoses

Coma or confusion of unknown origin

Neonatal resuscitation

Contraindications: Known sensitivity

Precautions: Pregnancy

Rapid administration or high dose may cause

Side Effects:

Dose:

Adult:

Pediatric:

Severe withdrawal effects in patients dependent on narcotics.

Abrupt administration may cause nausea, vomiting, sweating, tachycardia, increased blood pressure, seizures

0.4 mg increments to a max of 2 mg SIVP, titrated to achieve satisfactory respiratory effort. Monitor EtCO<sub>2</sub>.

If unable to obtain IV access consider IN administration using a one time 4 mg Narcan nasal spray or Narcan 0.4 -2 mg titrated to respiratory effort.

Refer to Broselow Tape

Neonatal resuscitation: 0.1 mg/kg

Routes: IV, IO, ET, IN

08/05/2026

NITROGLYCERIN

Indications: Chest pain

CHF

Contraindications: Systolic BP <100

Precautions:

Side Effects:

Dose:

Route:

Pulse rate < 50 and >140

Known sensitivity

Cerebral hemorrhage

Pregnancy

Headache

Postural hypotension

Flushing

Dizziness

Chest Pain: 0.4 mg every five minutes until chest pain is relieved or systolic blood pressure becomes <100 or 3 doses are given.

\*Patients taking NTG on a regular basis, may be given NTG prior to obtaining IV access.

CHF: 0.4 mg. May be repeated every five minutes until distress is relieved or systolic blood pressure <100 or 3 doses.

Sublingual

05/23/2025

NITROUS OXIDE

Indications: Pain associated with isolated orthopedic extremity trauma

Contraindications: Multi-systems trauma

Chest trauma

Head trauma

COPD

Dose: Self-administered

Route: Inhalation

05/23/2025

SODIUM BICARBONATE

Indications: Near drowning with prolonged submersion

Tricyclic antidepressant overdose with wide complex tachycardia

VF/Pulseless VT

Contraindications: Respiratory Alkalosis

Precautions:

Dose:

Route:

Metabolic Alkalosis

May cause alkalosis or tetany

1 meq/kg

Must contact medical control prior to administering for  
tricyclic antidepressant overdoses that is not in cardiac  
arrest.

IV, IO

05/23/2025

TERBUTALINE SULFATE

INDICATIONS:

Moderate to severe bronchial asthma

Reversible bronchospasm in COPD

ADMINISTRATION:

SQ only

DOSAGE:

ADULT:

0.25 mg (0.25 ml)

PEDIATRIC:

0.01 mg/kg Maximum Dose = 0.25 mg

THERAPEUTIC EFFECTS:

- Bronchodilator through selective beta 2 stimulation
- Relatively little effect on heart rate
- Relaxes bronchial smooth muscle

RELATIVE CONTRAINDICATIONS:

- Angina
- Dysrhythmias
- Hypertension
- Hypersensitivity to the drug

SIDE EFFECTS:

- Palpitations
- Tremor
- Drowsiness
- Headache
- Sweating
- Nausea
- Muscle cramping

SPECIAL NOTES / RESTRICTIONS:

- May repeat dosage in 15-30 minutes
- Do not give with other sympathomimetic drugs



04/01/2026

TRANEXAMIC ACID (TXA)

Indications:

Contraindications:

Side Effects:

Dose:

Route:

Classification:

Adult patients within 3 hours of traumatic injury with suspected hemorrhage.

Signs/Symptoms of shock (hypotension, tachycardia, pallor, etc.)

Pediatric patients

Isolated traumatic brain injury

Known thrombotic disease (DVT, PE)

Known allergy

Dizziness

Nausea

Hypotension (If administered too fast)

Visual disturbance (blurred vision, color distortion)

1 gram in 100mL of NS over 10 minutes

IV drip

Anti-Fibrinolytic

05/23/2025

VERSED (MIDAZOLAM)

Indications: Seizures

Sedation

Excited Delirium

Contraindications: Known sensitivity to the drug or benzodiazepines

Shock

Pregnancy

Precautions: COPD

CHF

Side Effects: Respiratory depression

Apnea

Hypotension

Nausea/vomiting

Dose: Seizures: seizure activity is present, 5-10 mg SIVP, may repeat in 10-15 minutes. You may also consider IN administration adult 5 mg in each nare.

Sedation: 2-4 mg may repeat 2 mg increments up to a total dose of 10 mg. If IV access is unavailable consider IN administration at a dose of 2-4mg.

Excited Delirium: 5 mg IM may repeat up to a total dose of 10 mg. If IV access is unavailable consider IN administration. Contact Medical control immediately for higher dosing. Be prepared to aggressively maintain airway.

Pediatric Dose: Seizures: 0.2 mg/kg IN, not to exceed 10 mg.

Sedation: 0.05 mg/kg slow IV (maximum initial dose 2-4 mg).

0.2 mg/kg IN, not to exceed 10 mg.

Route: IV, IO, IM, IN

05/23/2025

XOPENEX

Indications: Asthma/COPD

Contraindications: Known sensitivity

Heart rate greater than 150

Side Effects: Increased heart rate

Increased blood pressure

Dose: 1.25 mg nebulized. May repeat to a total of 3 treatments or heart rate exceeds 150 bpm.

Pediatric Dose: For children 1.25 mg mixed in 3 ml of NS nebulized.

Route: Nebulized

05/23/2025

ZOFRAN (ONDANSETRON)

Classification:

Anitemetic

Actions/Pharmacology:

Blocks the effects of serotonin at 5-HT3 receptor sites (selective antagonist) located in vagal nerve terminals and the chemoreceptor trigger zone in the CNS.

Indications:

Nausea and vomiting, regardless of mechanism.

Contraindications:

Age < 4 years

Parkinson's patients receiving Apomorphine

Side and Adverse effects:

Headache

Sedation

Constipation

Diarrhea

Transient blindness

May cause transient dizziness after IV administration

May cause prolonged QT interval and/or Torsade de Pointes

May cause transient oculogyric crisis and dystonic reactions

Adult Administration:

4mg IM or orally dissolving tablet (ODT)

4mg slow IVP over 2 to 5 minutes

Must contact medical control for additional doses.

Pediatric Administration:

Must be 4 years of age and older

2mg IM or ODT

2mg slow IVP over 2 to 5 minutes

Must contact medical control for additional doses.

Interactions:

When taken with apomorphine, may cause hypotension and ALOC

Precautions:

Tablet should not be handled with moist or wet hands

May cause prolonged QT interval and/or Torsade de Pointes

Patients receiving IV administration should be placed on cardiac monitor

Special Considerations:

Tablet should be placed onto tongue and allowed to dissolve.

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Reference Pages

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Tricyclic Antidepressant List, Common

The TCAs include the following agents which are predominantly serotonin and/or norepinephrine reuptake inhibitors:

- \* Amitriptyline (Tryptomer, Elavil)
- \* Amitriptylinexide (Amioxid, Ambivalon, Equilibrin)
- \* Butriptyline (Evadyne)
- \* Clomipramine (Anafranil)
- \* Demexiptiline (Deparon, Tinoran)
- \* Desipramine (Norpramin, Pertofrane)
- \* Dibenzepin (Noveril, Victoril)
- \* Dimetacrine (Istonil, Istonyl, Miroistonil)
- \* Dosulepin/Dothiepin (Prothiaden)
- \* Doxepin (Adapin, Sinequan)
- \* Imipramine (Tofranil, Janimine, Praminil)
- \* Imipraminoxide (Imiprex, Elepsin)
- \* Lofepramine (Lomont, Gamanil)
- \* Melitracen (Deanxit, Dixeran, Melixeran, Trausabun)
- \* Metapramine (Timaxel)
- \* Nitroxazepine (Sintamil)
- \* Nortriptyline (Pamelor, Aventyl, Norpress)
- \* Noxiptiline (Agedal, Elronon, Nogedal)
- \* Pipofezine (Azafen/Azaphen)
- \* Propizepine (Depressin, Vagran)
- \* Protriptyline (Vivactil)
- \* Quinupramine (Kevopril, Kinupril, Adeprim, Quinuprine)

As well as the following atypical compounds:

- \* Amineptine (Survector, Maneon, Directim) - Norepinephrine-dopamine reuptake inhibitor
- \* Iprindole (Prondol, Galatur, Tetran) - 5-HT<sub>2</sub> receptor antagonist
- \* Opipramol (Insidon, Pramolan, Ensidon, Oprimol) - receptor agonist
- \* Tianeptine (Stablon, Coaxil, Tatinol) - Selective serotonin reuptake enhancer
- \* Trimipramine (Surmontil) - 5-HT<sub>2</sub> receptor antagonist

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#### Calcium Channel Blockers

Calcium channel blockers are called CCBs for short. They are also called calcium channel antagonists. CCBs are the most widely prescribed medicines in the United States today. They can be used alone or in a combination pill with an ACE inhibitor. When are calcium channel blockers prescribed?

In addition to treating those with high blood pressure, calcium channel blockers are also used for people who have heart disease. Because they allow the blood vessels that supply the heart to open wider, more oxygen-rich blood can get to the heart. They are also used to treat arrhythmia.

#### Blockers

The following table lists some common brand and generic names for calcium channel blockers.

| Brand | Generic |
|-------|---------|
|-------|---------|

|              |                     |
|--------------|---------------------|
| Adalat CC    | Nifedipine          |
| Calan SR     | verapamil HCl       |
| Cardene      | nicardipine HCl     |
| Cardene SR   | nicardipine HCl     |
| Cardizem CD  | diltiazem HCl       |
| Cardizem SR  | diltiazem HCl       |
| Covera-HS    | verapamil HCl       |
| Dilacor XR   | diltiazem HCl       |
| DynaCirc     | Isradipine          |
| DynaCirc CR  | Isradipine          |
| Isoptin SR   | verapamil HCl       |
| Nimotop      | Nimodipine          |
| Norvasc      | amlodipine besylate |
| Plendil      | Felodipine          |
| Procardia XL | Nifedipine          |
| Sular        | Nisoldipine         |
| Tiamate      | diltiazem HCl       |
| Tiazac       | diltiazem HCl       |
| Vascor       | bepidil HCl         |
| Verelan      | verapamil HCl       |
| Verelan PM   |                     |

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MISSOURI APPROVED DNR ORDER



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Out of Hospital DNR FAQ

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PEDIATRIC EMERGENCY FORM

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TPOPP for EMS

What is TPOPP?

Transportable Physician Orders for Patient Preferences (TPOPP) is a program designed to improve the quality of care people receive at the end of life by translating patient/resident goals and preferences into medical orders for treatment.

What is the TPOPP form?

The TPOPP form is a bright pink piece of paper that can communicate patient preferences regarding life-sustaining medical treatment. When signed by a patient (or a recognized decision maker) and a physician licensed in Kansas or Missouri, it becomes a medical order. Providers at all sites of care should follow the medical orders.

What types of medical orders are on the form?

The form includes medical orders and patient preferences regarding:

CPR

Implementing comfort care

Defibrillation and cardioversion

Intubation and mechanical ventilation

Transport decisions

Who might have a TPOPP form?

TPOPP is designed for persons who have advanced progressive chronic illness or are terminally ill. EMS may see TPOPP forms for persons who:

Live independently;

Reside in a long-term care or assisted living facility;

Receive hospice care.

Does every person with advanced chronic or terminal illness have a form?

No. TPOPP is voluntary and based on the belief that an individual has the right to make his or her own health care decisions. TPOPP forms can be used by persons of any age with an advanced progressive illness or who is terminally ill.

How is a TPOPP form completed?

The TPOPP form is completed after having a conversation with the patient (or recognized decision maker) and a member of the healthcare team such as a physician, nurse, social worker or chaplain. When the patient (or recognized decision maker) and a physician sign the form it is complete. Both signatures are necessary. The signatures verify that the preferences on the form are consistent with the patient's values and treatment goals.

What do you do with a completed form?

The TPOPP form travels with the patient between care settings. The original form should accompany the patient and be presented to each provider (first responder, hospital or other care setting).

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Where is the form kept?

EMS personnel should look for or ask for the form. It should be conspicuously located:

In the front of the patient's medical chart at care facilities;

On the refrigerator, by a phone, or bedside when the patient resides at home;

The TPOPP form should go with the patient when the person is transferred from the hospital or health care setting.

How is the form used by EMS?

EMS should administer the level of treatment ordered in Section A and in Section B when they arrive on scene. EMS will transfer the individual to the emergency room if ordered or contact medical control for further instructions.

What sections apply to EMS providers?

Section A orders apply only when the person is in cardiopulmonary arrest. If the "Attempt Resuscitation/CPR" box is checked then CPR should be performed. If the "Do Not Attempt Resuscitation (DNAR/no CPR/Allow Natural Death)" box is checked, CPR should not be performed. Comfort measures are provided and CPR is not attempted.

Section B orders apply to emergency medical circumstances for a person who has a pulse and/or is breathing. These range from full treatment to comfort measures only. If all life-sustaining treatments are desired the "Full Treatment" box is checked. However, if the person desires some limitation, then either the "Comfort Measures Only" box or the "Limited Additional Interventions" box is checked. EMS first will administer the level of emergency medical services ordered and then contact medical control as needed.

Comfort care is always provided regardless of indicated level of EMS treatment.

If Attempt Resuscitation in Section A is checked, then Full Treatment in Section B is checked. Full resuscitation implies full treatment. Section D validates the order with the dated signatures of the physician and patient (or recognized decision maker).

When can EMS honor the TPOPP form?

EMS personnel can honor the TPOPP form in any situation in which a valid TPOPP form is presented for the patient requiring emergency medical care.

Will the person still have or need a signed Outside the Hospital Do Not resuscitate (DNR) form?

Your organization/agency will determine the specific policies associated with TPOPP.

Remember, that if DNAR is selected in Section A and the form is valid, you have a physician's order not to resuscitate.

Is a copy or faxed copy of the form acceptable?

Yes - A copy, faxed copy, or electronic representation of the TPOPP form is acceptable provided it has the person's name and signatures of the physician and patient (or recognized decision maker).

Where can I get more information?

Visit the website of the Center for Practical Bioethics at [www.practicalbioethics.org](http://www.practicalbioethics.org), send an e-mail to [TPOPP@practicalbioethics.org](mailto:TPOPP@practicalbioethics.org) or call the TPOPP Managing Director at 816-979-1366.

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TPOPP FORM

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#### Apneic Oxygenation

Prior to intubation apply a nasal cannula and flow at 8-10 LPM

In an experiment by Frumin et al., patients were pre-oxygenated, intubated, paralyzed, and placed on an anesthesia machine that provided 1.0 FiO<sub>2</sub> and no ventilations. These patients were maintained in this apneic state for between 18 and 55 min. none of these patients desaturated below 98%, despite being paralyzed and receiving no breaths. Although their CO<sub>2</sub> levels rose, their oxygenation was maintained due to apneic oxygenation. Oxygen was absorbed from the patients' alveoli by pulmonary blood flow; this established a gradient for the continued pull of oxygen from the endotracheal tube and anesthesia circuit. In another study, Teller et al. showed that pharyngeal insufflation with oxygen significantly extended the

time to desaturation during apnea.

Numerous studies on apneic oxygenation during brain death testing confirm that even without any respiratory effort, oxygen saturation can be maintained. If a continuous path of oxygen is maintained from the pharynx to the glottis during the apneic period of RSI, the patient will continue to oxygenate. This has led us to perform a jaw thrust in all high-risk patients during their apneic period. In some cases, we also place nasopharyngeal airways to augment the passage of oxygen. These techniques, combined with high-flow O<sub>2</sub> from a High flow cannula, NRB mask, NIV mask, or the facemask of a BVM, will allow continued apneic oxygenation.



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NIH Stroke Scale

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Restraint Checklist

Patient's Name: \_\_\_\_\_

Report Number: \_\_\_\_\_

1. Reason for restraint (check all that apply):

- ☐ Patient attempting to hurt self
- ☐ Patient attempting to hurt others
- ☐ Patient attempting to remove medical necessary devices
- ☐ Patient under arrest by Law enforcement (If yes then Law Enforcement Officer MUST accompany the patient and attending paramedic in the back of the ambulance.)

2. Attempted verbal assurance/redirection?

☐ Yes

☐ No

3. Attempted environmental modification?(i.e. remove patient from stressful environment)

☐ Yes

☐ No

4. Received medical control order for restraints?

☐ Yes

☐ No

5. Time and Type of restraint applied (circle all that apply):

Date: \_\_\_\_/\_\_\_\_/\_\_\_\_ Time: \_\_\_\_\_

Limb restraints: RUE, LUE, LLE, RLE

Chemical restraint: Yes No If Yes: Drug Used &

Dose: \_\_\_\_\_

6. Vital signs and extremity neurovascular exam should be taken every 10 min.

7. Transport Position (Patient should NOT be in prone position)

☐ Supine position for transport

☐ Lateral recumbent position for transport